

HEALTHINSURA360

**AI-Powered Health Insurance Management Platform
(FINAL YEAR PROJECT REPORT)**



**DEPARTMENT OF COMPUTER SCIENCES
COMSATS UNIVERSITY ISLAMABAD, WAH CAMPUS
WAH CANTT – PAKISTAN
SESSION 2022-2026**

HEALTHINSURA360

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A DISSERTATION SUBMITTED AS A PARTIAL FULFILLMENT OF THE
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DEPARTMENT OF COMPUTER SCIENCES
COMSATS UNIVESITY ISLAMABAD, WAH CAMPUS
WAH CANTT – PAKISTAN

SESSION 2022-2026

FINAL APPROVAL

The final approval page will be provided by the department after the final evaluation.

DEDICATION

We dedicate this project to Almighty Allah, whose grace and guidance have sustained us throughout this journey. Our deepest gratitude extends to our loving parents, whose unconditional support and sacrifices have paved the way for our achievements. We are eternally grateful to our esteemed supervisor, Ma'am Salma Rashid, for her mentorship, wisdom, and unwavering belief in our potential. We sincerely thank our dedicated teachers, who have nurtured our growth and inspired us to strive for excellence.

ACKNOWLEDGEMENT

Our deepest gratitude goes to Almighty Allah, whose boundless mercy and guidance has sustained us throughout this project and our academic journey.

We are eternally grateful to our parents, whose love and sacrifices have made our achievements possible. We extend heartfelt thanks to our teachers, especially Ma'am Salma Rashid, for her unwavering support, mentorship, and Encouragement. Her guidance has been invaluable, and we carry her lessons With us.

We would also like to thank our fellow students and friends who provided Support and encouragement throughout this project.

PROJECT BRIEF

PROJECT NAME	HEALTHINSURA360
ORGANIZATION NAME	COMSATS UNIVERSITY WAH CAMPUS
OBJECTIVE	TO SIMPLIFY HEALTH INSURANCE PURCHASE MANAGEMENT, AND CLAIM PROCESSING USING AI- POWERED RECOMMENDATION AND DIGITAL WORKFLOWS
UNDERTAKEN BY	KAYANAT MUGHAL (FA22-BCS-007) QURATULAIN AZHAR (FA21-BCS-041)
SUPERVISED BY	MA'AM SALMA RASHID
STARTED ON	1 SEPTEMBER 2025
COMPLETED ON	2 MAY 2026
COMPUTER USED	intel core i5 12 th Generation
SOURCE LANGUAGE	JAVASCRIPT, PYTHON
OPERATING SYSTEM	WINDOWS / LINUX
TOOLS USED	VSCODE, REACT JS, NODE JS, EXPRESS JS, POSTGRESQL, FIGMA

ABSTRACT

HealthInsura360 is an AI-powered web-based health insurance management platform designed to simplify the process of purchasing, understanding, and managing health insurance. The system integrates policy comparison, AI-driven recommendations, claim processing, and secures digital payments into one unified ecosystem. It serves four distinct stakeholders: Users (Policyholders), Agents, Admins, and Hospitals. Key features include JWT-based authentication, real-time chatbot support, digital claim submission and tracking, secure payment gateway integration, and comprehensive analytics dashboards.

Traditional health insurance portals often suffer from lack of personalization, confusing interfaces, and manual claim processing. HealthInsura360 addresses these deficiencies by integrating AI-backed policy recommendations and an intelligent chatbot that assists users in real-time. The system reduces user confusion, enhances policy accessibility, increases transparency in claim processing, and modernizes the insurance ecosystem by integrating artificial intelligence, digital payments, analytics, and secure identity management.

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1: INTRODUCTION

The purpose of this chapter is to present a detailed description of HealthInsura360. It will explain the system's introduction in detail, the background of the system, its objectives, and the significance of the system.

1.1 System Introduction

HealthInsura360 is an AI-powered web-based health insurance management platform designed to simplify the process of purchasing, understanding, and managing health insurance. The system integrates policy comparison, AI-driven recommendations, claim processing, and secures digital payments into one unified ecosystem. It serves multiple stakeholders including Users (Policyholders), Admins, Agents, and Hospitals, providing specialized tools to efficiently perform their required tasks.

The platform leverages artificial intelligence to provide personalized policy recommendations based on user profiles, preferences, and historical data. An intelligent chatbot assists users in real-time, answering queries about policies, claims, and system usage. The system also facilitates cashless claim submissions from hospitals, agent-based claim verification, and admin-level oversight of the entire ecosystem.

1.2 Background of the System

Existing health insurance portals often suffer from lack of personalization, confusing interfaces, and manual or slow insurance claim processes. Users face difficulties understanding policy terminology, comparing different plans, and managing multiple insurance documents. Key problems identified in existing systems include:

- **Limited Personalization:** Most portals offer generic policy recommendations without considering individual user needs and medical history.
- **Complex Claim Processes:** Manual document submission and lengthy verification times create frustration for policyholders.
- **Poor Communication:** Lack of real-time updates on claim status and policy renewals.

- **Fragmented Ecosystem:** Hospitals, agents, and users operate in silos without integrated workflows.
- **Security Concerns:** Inadequate authentication mechanisms and data protection measures.

HealthInsura360 aims to address these deficiencies by integrating AI-backed policy recommendations and a chatbot that assists users in real-time, creating a seamless, transparent, and efficient insurance management experience.

1.3 Objectives of the System

The main objectives of developing HealthInsura360 are:

- i. **Simplify Health Insurance Management:** Provide an intuitive platform for users to explore, purchase, and manage health insurance policies.
- ii. **Provide Personalized Policy Recommendations:** Implement AI algorithms that analyze user profiles and suggest the most suitable policies.
- iii. **Enable Digital Claim Processing:** Allow users and hospitals to submit claims online with document uploads and real-time tracking.
- iv. **Facilitate Transparent Claim Approvals:** Establish a clear workflow where agents review claims and admins maintain oversight.
- v. **Offer Secure Payment Experience:** Integrate payment gateways with JWT-based authentication for secure transactions.
- vi. **Provide Analytics and Reporting Tools:** Deliver comprehensive dashboards for admins to monitor system performance and make data-driven decisions.
- vii. **Enhance Communication:** Implement notification systems (email/SMS) for policy updates, claim status changes, and renewal reminders.
- viii. **Implement Admin Approval Workflow:** Require admin verification for hospital registrations before granting system access.
- ix. **Auto-generate Agent Credentials:** Automatically generate unique license numbers for registered agents.

x. **Smart Policy Filtering:** Automatically hide policies already owned by users when browsing available plans.

1.4 Significance of the System

HealthInsura360 holds the potential to transform the health insurance experience for all stakeholders:

For Users (Policyholders):

- Simplified policy discovery through AI recommendations
- Transparent claim tracking and faster processing
- 24/7 assistance through AI chatbot
- Secure digital payments and policy document access

For Insurance Agents:

- Streamlined claim review and verification workflows
- Centralized view of assigned customers
- Commission tracking and reporting
- Efficient document validation tools

For Hospitals:

- Digital cashless claim submission
- Quick patient policy verification
- Medical document upload and management
- Real-time claim status tracking

For Admins:

- Complete system oversight and control
- Comprehensive analytics and audit logs
- Efficient user and policy management
- Data-driven decision-making capabilities

2: REQUIREMENT SPECIFICATIONS

The purpose of this chapter is to translate the functional requirements of HealthInsura360 into technical guidelines for implementation. It provides detailed descriptions of product scope, product description, functional requirements, behavioral requirements, external interfaces, and non-functional requirements.

2.1 Product Scope

2.1.1 Functions

HealthInsura360 provides the following core functions:

- i. **User Account Management:** Registration, login, profile updates, and document uploads with JWT-based authentication.
- ii. **AI-Driven Policy Recommendations:** Personalized policy suggestions based on user demographics, health profiles, and preferences.
- iii. **AI Chatbot Support:** Real-time conversational assistance for policy inquiries and claim guidance.
- iv. **Policy Management:** Browsing, filtering, purchasing, and renewal of insurance policies with digital certificates.
- v. **Claim Management:** Submission, document upload, tracking, and processing of reimbursement and cashless claims.
- vi. **Agent Tools:** Customer management, claim review, approval/rejection workflows, and commission tracking.
- vii. **Hospital Portal:** Patient policy verification, cashless claim submission, treatment details entry, and document upload.
- viii. **Admin Dashboard:** User management, policy CRUD operations, analytics, audit logs, and system monitoring.

ix. **Payment Processing:** Secure payment gateway integration with receipt generation and invoice downloads.

x. **Notification System:** Email and SMS alerts for policy events, claim updates, and renewal reminders.

2.1.2 Boundaries

HealthInsura360 is specifically focused on health insurance management. It does NOT include:

- i. Offline claim processing (all claims are digital)
- ii. Real banking integration (test mode gateways are used)
- iii. Real-time third-party insurance verification
- iv. Integration with external hospital management systems

2.2 Product Description

2.2.1 Product Perspective

HealthInsura360 is a standalone web-based product built using modern web technologies, AI modules, and secure backend services. It interacts with payment gateways, AI models, email/OTP services, and hospital systems

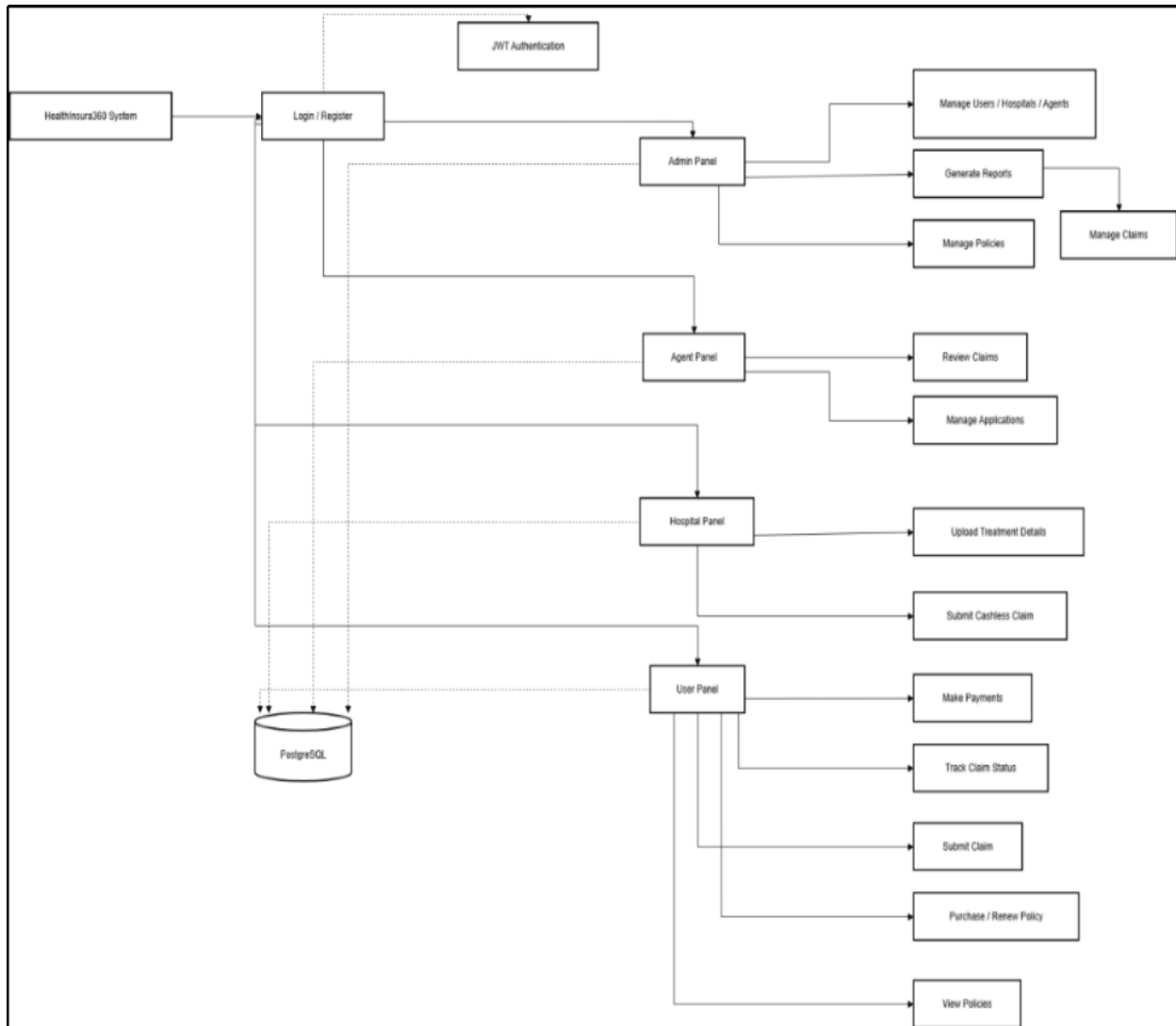


Figure 2. 1: System Design

The system consists of multiple tightly integrated modules:

Table 2. 1: Modules

User Module	Registration, profile, policy viewing, claim submission
Agent Module	Customer management, claim review, commission tracking
Admin Module	System oversight, user management, analytics
Hospital Module	Patient verification, cashless claims, document upload
AI Module	Policy recommendations, chatbot
Payment Module	Secure transaction processing
Authentication Module	Uses JWT to securely verify user identities and manage role-based access.

2.2.2 Product Functionality

Following are the functional capabilities of HealthInsura360:

User Functions:

- Account registration and login
- Profile management
- View and compare policies
- Get AI-based policy recommendations
- Chat with AI chatbot
- Purchase and renew policies
- Submit reimbursement claims
- Upload claim documents
- Track claim status
- Make secure payments
- Download policy documents
- Receive notifications

Agent Functions:

- Login and dashboard access
- View assigned customers
- Manage customer applications
- Review, approve, or reject claims
- Track commissions

Hospital Functions:

- Login to hospital portal
- Search patient policies
- Enter treatment details
- Upload medical documents
- Submit cashless claims
- Track claim status
- Generate claim reports

Admin Functions:

- Login to admin panel
- Manage user accounts (all types)
- Manage policies (CRUD operations)
- Generate reports
- View analytics dashboard
- Handle complaints
- Manage claims (oversight)
- View audit logs
- Manage payment transactions
- Review agent commissions
- Verify hospital network

2.2.3 Users and Characteristics

HealthInsura360 will have four distinct user types:

2.2.3.1 User (Customer / Policyholder)

- **Description:** Regular customers who register, purchase policies, submit claims
- **Frequency of Use:** Occasional to moderate
- **Technical Expertise:** Basic
- **Privilege Level:** Lowest (only own profile, policies, claims)
- **Tasks They Perform:**
 - Register/Login
 - View available policies
 - Purchase or renew policies
 - Submit claims with documents
 - Track claim status
 - Receive notifications
 - Chat with chatbot

2.2.3.2 Agent

- **Description:** Licensed personnel helping users with onboarding and claims
- **Frequency of Use:** High
- **Technical Expertise:** Moderate
- **Privilege Level:** Medium-high (access to assigned customers)
- **Registration:** Auto-generated license number upon successful registration
- **Tasks They Perform:**
 - Register and receive auto-generated license number
 - Login to access agent dashboard
 - View Assigned Customers
 - Verify customer identity/documents
 - Manage customer applications
 - Review incoming claims
 - Approve/reject claims submission
 - Track commissions and performance metrics

2.2.3.3 Admin

- **Description:** Highest-level user managing entire system
- **Frequency of Use:** High to moderate
- **Technical Expertise:** High
- **Privilege Level:** Highest (full system access)
- **Tasks They Perform:**
 - Manage all user accounts
 - Add/modify/delete policies
 - View system analytics
 - Review audit logs
 - Handle complaints
 - Monitor payment transactions

2.2.3.4 Hospital

- **Description:** Hospital representatives who validate insurance and submit cashless claims
- **Frequency of Use:** Moderate
- **Technical Expertise:** Low to moderate
- **Privilege Level:** Medium (claim-related functions only)
- **Registration Process:**
 - Hospital submits registration documents

- Admin reviews and approves/rejects
- Upon approval, email notification sent with setup link
- Hospital verifies token and sets password
- Login access granted to dashboard
- **Tasks They Perform:**
 - Register by submitting hospital documents (pending status)
 - Wait for admin approval
 - Receive email notification with setup link upon approval
 - Verify token and set password
 - Login to hospital dashboard
 - Verify patient insurance details
 - Submit cashless claims for patient treatment
 - Upload medical documents
 - Track submitted claim status

2.2.4 Operating Environment

HealthInsura360 operates in a fully web-based environment, accessible through modern web browsers.

Software Tools:

- Visual Studio Code (VS Code)
- Git & GitHub (version control)

Programming Languages & Frameworks:

- React.js (Frontend)
- Node.js (Backend runtime)
- Express.js (Backend framework)
- PostgreSQL (Database)

Operating System:

- Windows 10 / 11 / Linux / macOS

Hardware Specifications:

- **Processor:** Intel Core i5 (6th Gen+)
- **RAM:** 8 GB (16 GB recommended)

- **Storage:** 256 GB SSD
- **Internet:** 2 Mbps (10+ Mbps recommended)

2.3 Specific Requirements

2.3.1 Functional Requirements

- User Management

Table 2. 2: Functional Requirement 1

Requirement ID	Requirement Description
REF-001	Users can register, login, reset passwords securely, and access role-based accounts (User, Agent, Hospital, Admin).
REF-002	Users can view and update personal details, upload documents, and track purchased policies and claim history.
REF-003	System ensures account security using standard email-and-password authentication

- Policy Management

Table 2. 3: Functional Requirement 2

Requirement ID	Requirement Description
REF-004	Users can browse, filter, purchase, and renew insurance policies with payment integration and digital certificates.
REF-005	Admin can add, edit, remove, and manage visibility of policies.

REF-006	System automatically filters out policies already owned by user when browsing available plans
---------	---

- Claim Management

Table 2. 4: Functional Requirement 3

Requirement ID	Requirement Description
REF-007	Users or hospitals can submit claims with supporting documents.
REF-008	Agents review, verify, approve, or reject claims, with notifications sent to users.
REF-009	Users and hospitals can track claim status, verification progress, and system maintains audit logs.
REF-010	Users can upload claim documents including medical bills and prescriptions
REF-011	Users can upload profile picture during profile management

- Agent Management

Table 2. 5: Functional Requirement 4

Requirement ID	Requirement Description
REF-012	Agents can review applications, verify documents, approve/reject applications, and track commissions.

- Hospital Interaction

Table 2. 6: Functional Requirement 5

Requirement ID	Requirement Description
-----------------------	--------------------------------

REF-013	Hospitals can verify policies, submit claims on behalf of patients, upload documents, and monitor claim status.
---------	---

- Admin Management

Table 2. 7: Functional Requirement 6

Requirement ID	Requirement Description
REF-014	Admin can manage roles, permissions, monitor system activities, and reset accounts.
REF-015	Admin can generate and export reports (claims, policies, agents, hospitals) and manage content and notifications.
REF-016	Admin can view list of pending hospital registrations
REF-017	Admin can approve or reject hospital registration requests
REF-018	System sends email notification to hospital upon approval with secure setup link

- Payment & Billing

Table 2. 8: Functional Requirement 7

Requirement ID	Requirement Description
REF-019	System processes payments securely, validates transactions, generates receipts, and allows invoice downloads.

- Notification & Communication

Table 2. 9: Functional Requirement 8

Requirement ID	Requirement Description
----------------	-------------------------

REF-020	System sends notifications (email/SMS) for policy and claim events.
REF-021	Users can view, mark as read, and manage notifications in real-time

2.3.2 Behavioral Requirements

Use Case Diagram

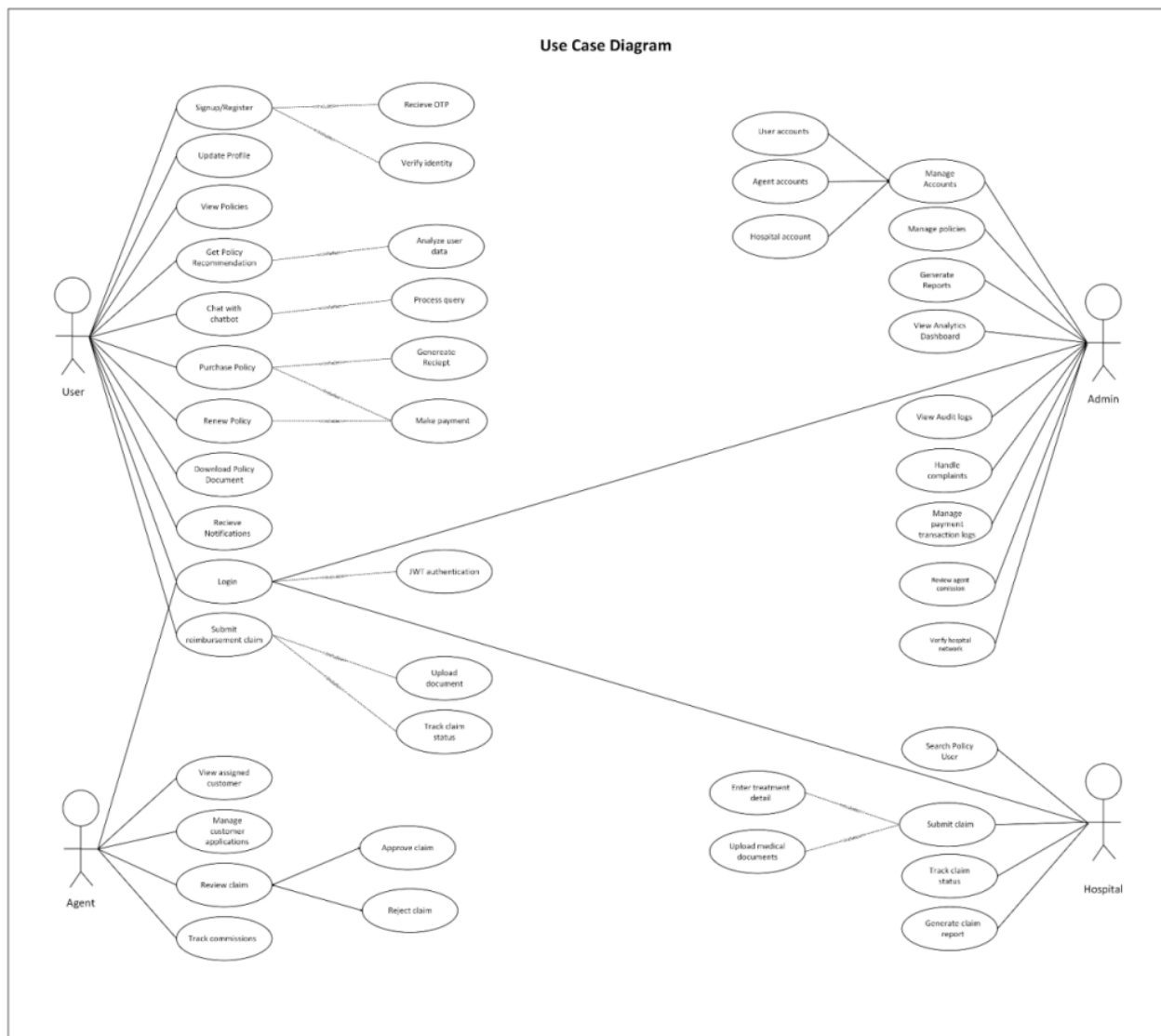


Figure 2. 2: Use Case Diagram

2.3.3 External Interfaces

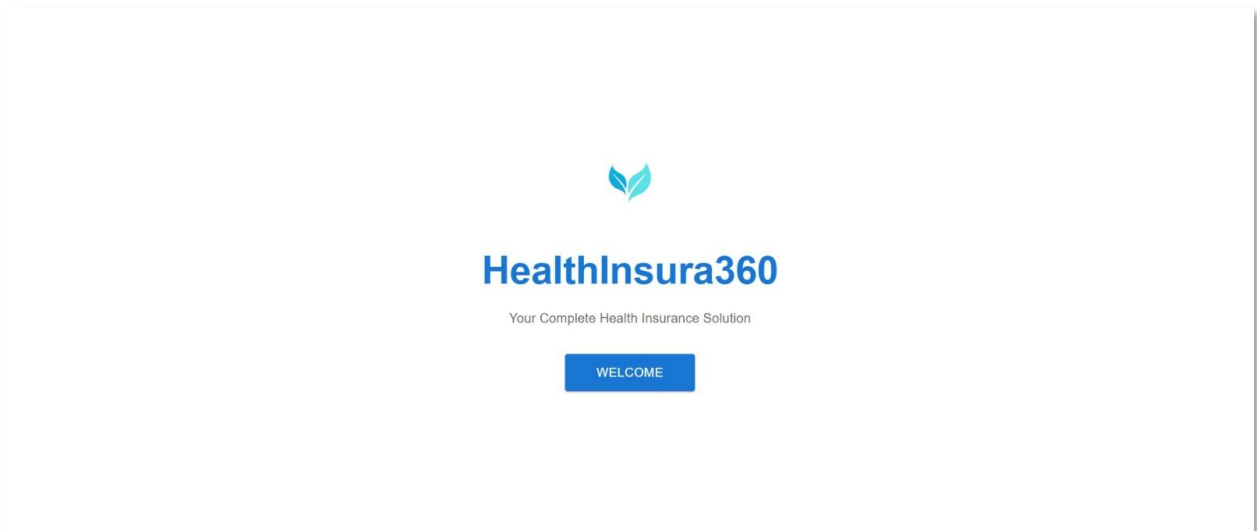


Figure 2. 3: Landing Page

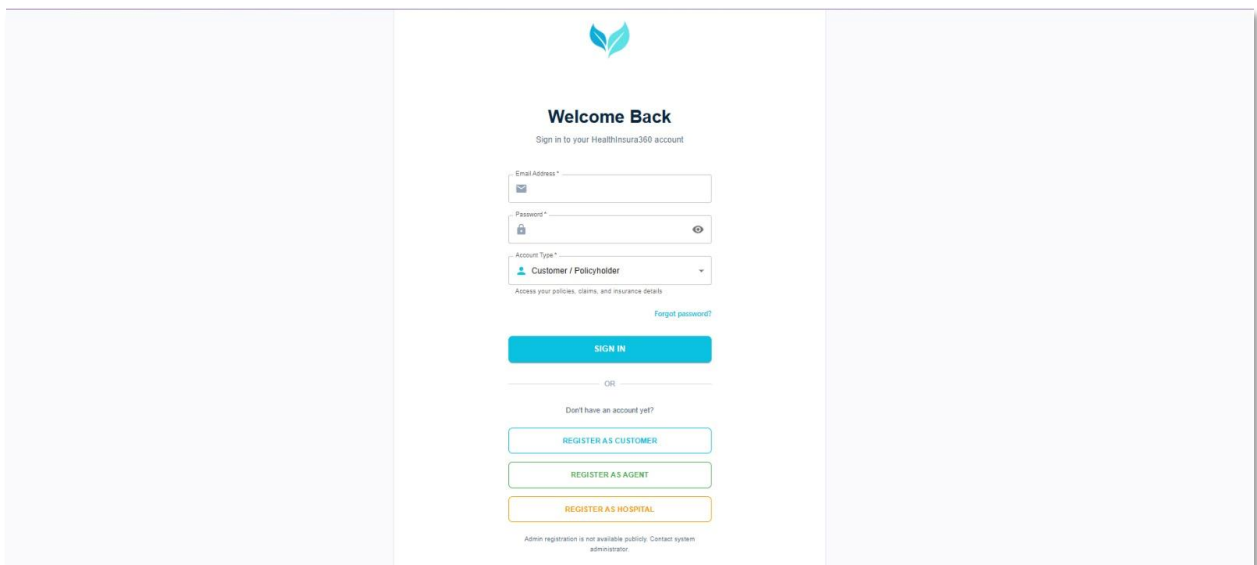
The image shows the login screen of HealthInsura360. At the top center is the same two-leaf logo. Below it, the heading "Welcome Back" is in bold black text, followed by the subtext "Sign in to your HealthInsura360 account". The login form includes three input fields: "Email Address" with an envelope icon, "Password" with a lock icon and a toggle eye icon, and "Account Type" with a dropdown menu currently showing "Customer / Policyholder" and a person icon. Below the dropdown is a link that says "Access your policies, claims, and insurance details" and another link "Forgot password?". A large blue "SIGN IN" button is positioned below the form. Underneath the button is a horizontal line with the word "OR" in the center. Below this line is the text "Don't have an account yet?". There are three registration buttons: "REGISTER AS CUSTOMER" (blue), "REGISTER AS AGENT" (green), and "REGISTER AS HOSPITAL" (orange). At the very bottom, in small text, it says "Admin registration is not available publicly. Contact system administrator."

Figure 2. 4: Login Screen

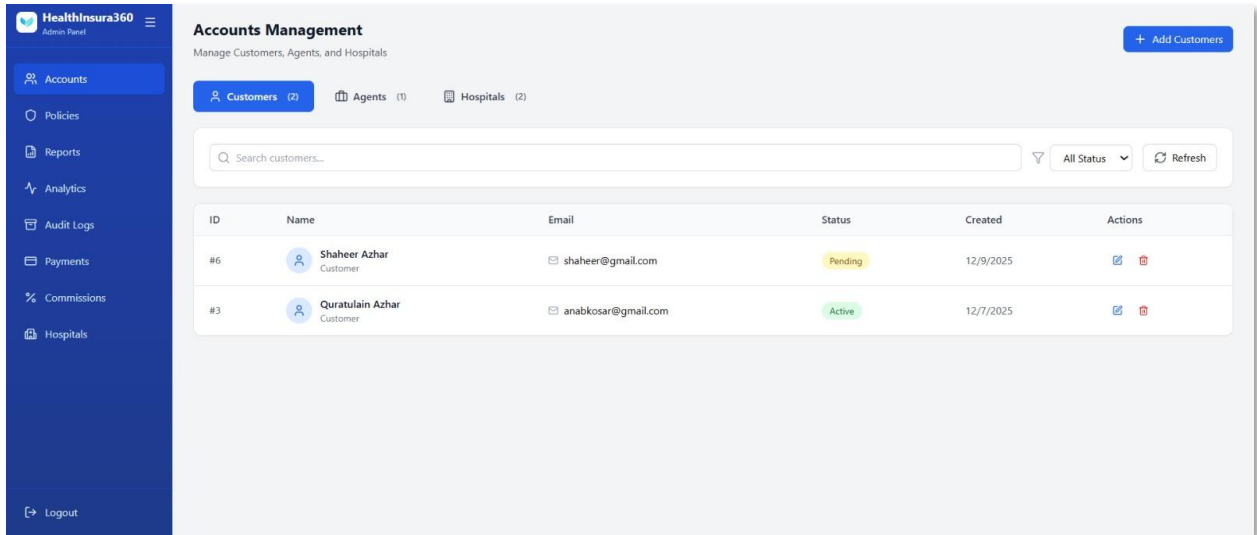


Figure 2. 5: Admin Screen

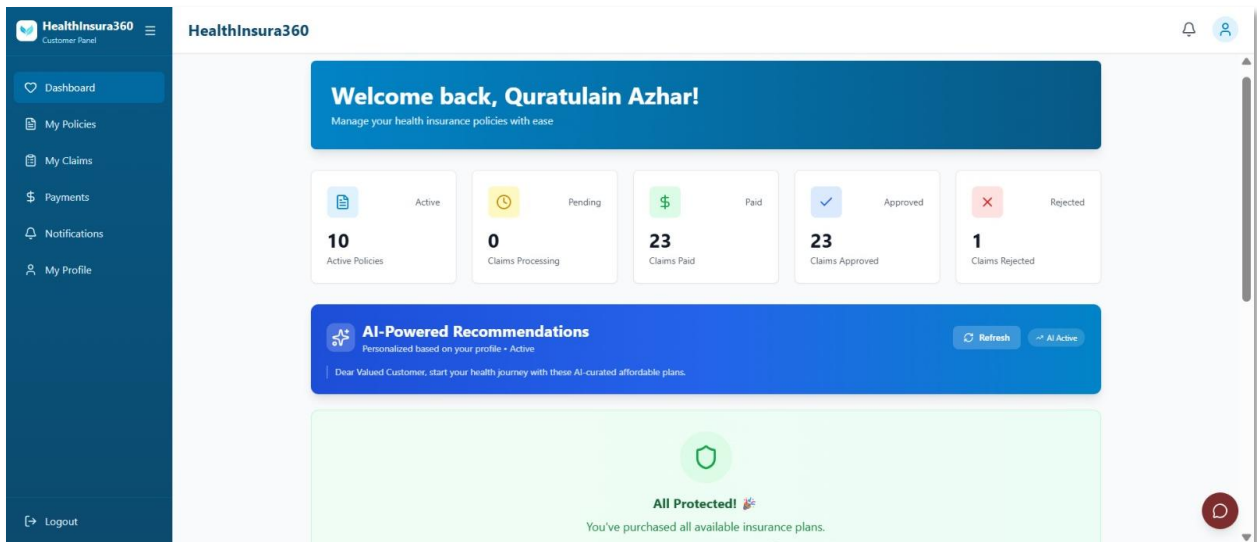


Figure 2. 6: Customer Screen

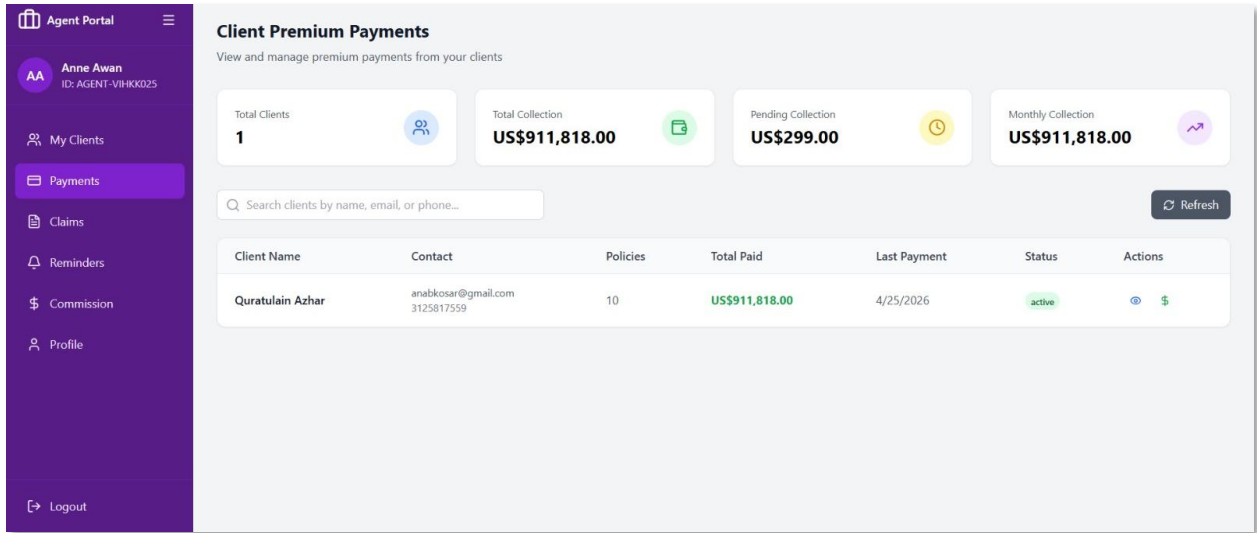


Figure 2. 7: Agent Screen

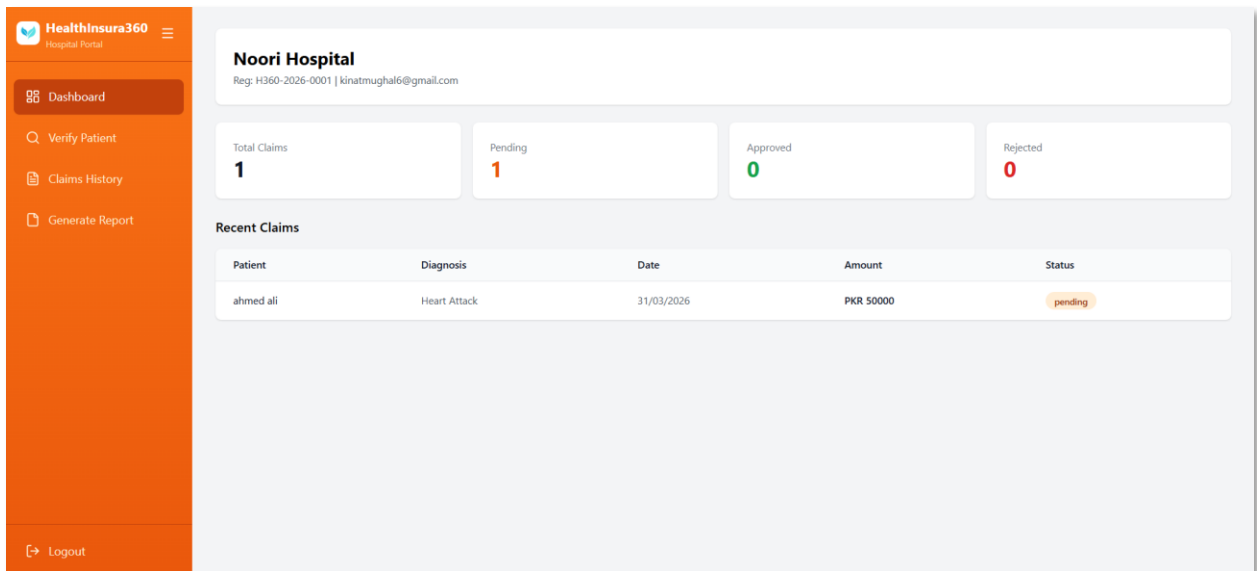


Figure 2. 8: Hospital Screen

2.4 Non-Functional Requirements

2.4.1 Performance Requirements

Table 2. 10: Performance Requirements

Requirement ID	Requirement Description
P.1	Claim submission and policy purchase confirmation within 5 seconds
P.2	Page loading within 3 seconds
P.3	Support 200 concurrent users without performance degradation
P.4	System availability 99.9% uptime

2.4.2 Safety and Security Requirements

Safety Requirements:

- S.1: Transaction logging to prevent financial data loss
- S.2: Full system restores with maximum 4-hour data loss
- S.3: Critical admin functions require re-authentication

Security Requirements:

- SEC.1: JWT-based authentication for all actors
- SEC.2: TLS 1.2+ for data in transit, encryption at rest
- SEC.3: Passwords stored with crypt hashing
- SEC.4: Session timeout after 20 minutes of inactivity

2.4.3 Software Quality Attributes

- **Reliability:** 99.9% uptime with redundant infrastructure
- **Maintainability:** Critical bugs fixed within 48 hours
- **Usability:** Complete purchase/claim within 10 minutes on first attempt
- **Portability:** OS-independent, containerized deployment

3: DESIGN SPECIFICATIONS

The purpose of this chapter is to present the design views of HealthInsura360 using UML diagrams. It includes composite viewpoint (package and deployment diagrams), logical viewpoint (class diagram), information viewpoint (ER diagram), interaction viewpoint (sequence diagrams), state dynamics viewpoint (state machine diagram), and algorithm viewpoint (pseudocode).

3.1 Introduction

This chapter presents the design views of HealthInsura360 using UML diagrams including Package Diagrams, Deployment Diagram, Class Diagram, Entity Relationship Diagram, Sequence Diagrams, and State Machine Diagram.

3.2 Composite Viewpoint

3.2.1 Package Diagram (Logical)

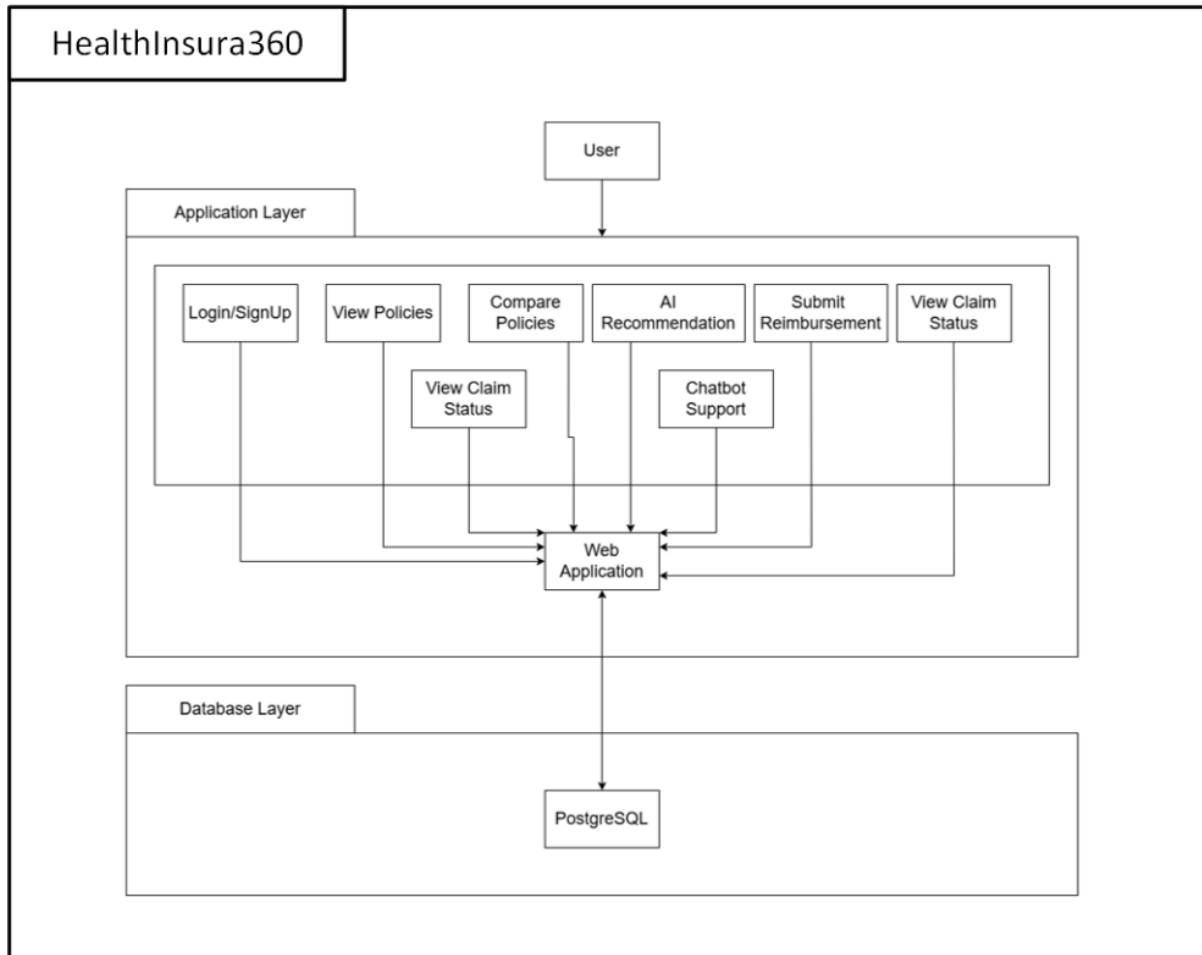


Figure 3. 1: User Package Diagram

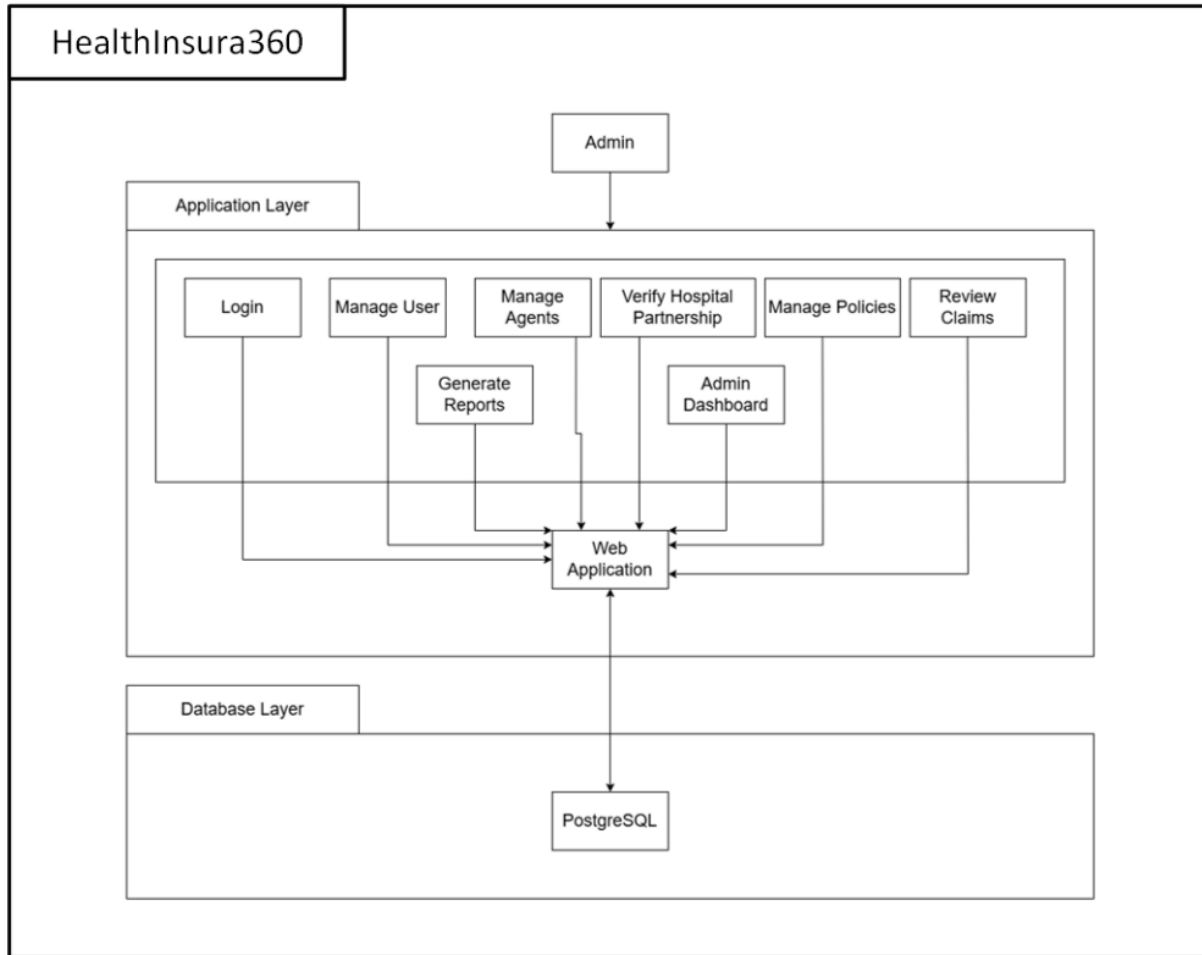


Figure 3. 2: Admin Package Diagram

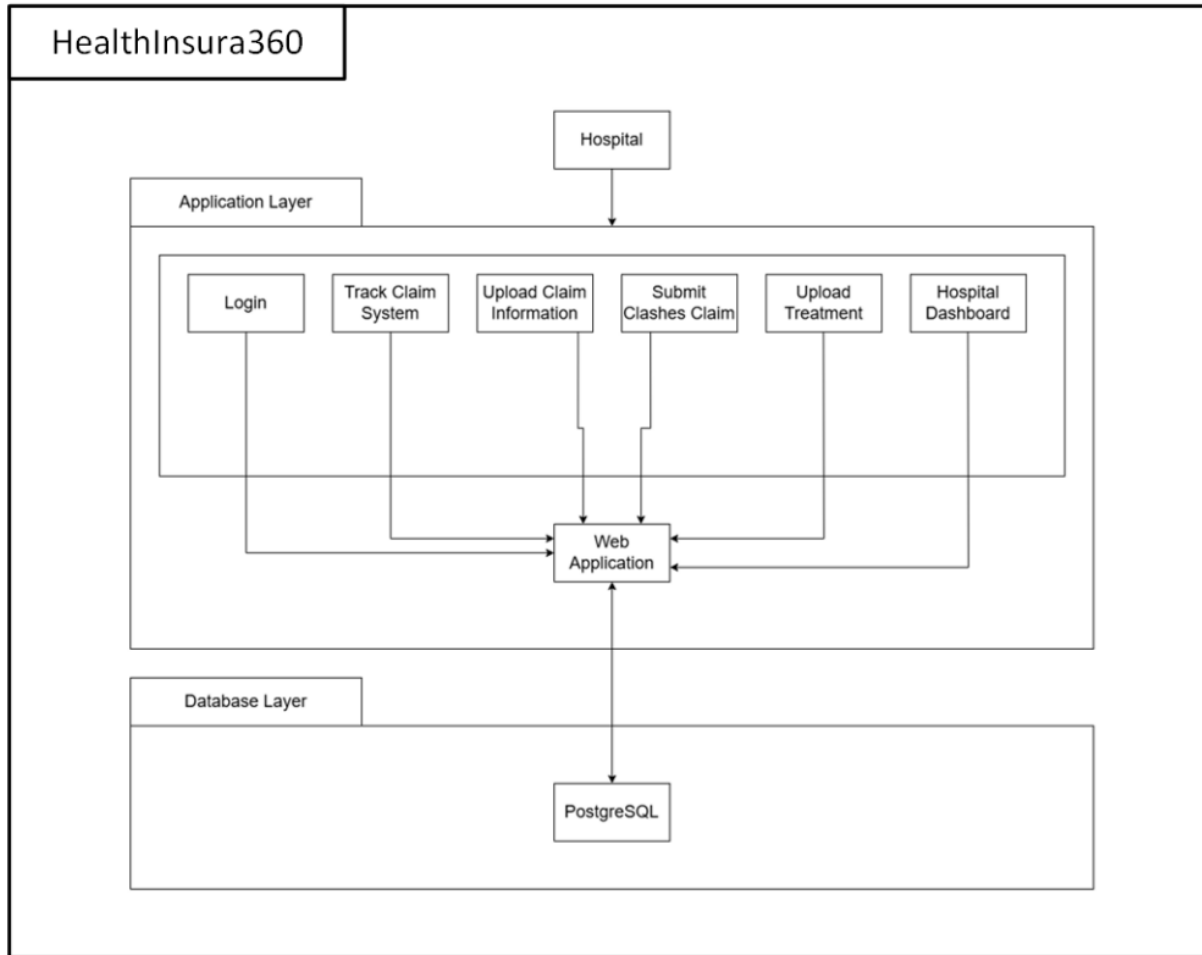


Figure 3. 3: Hospital Package Diagram

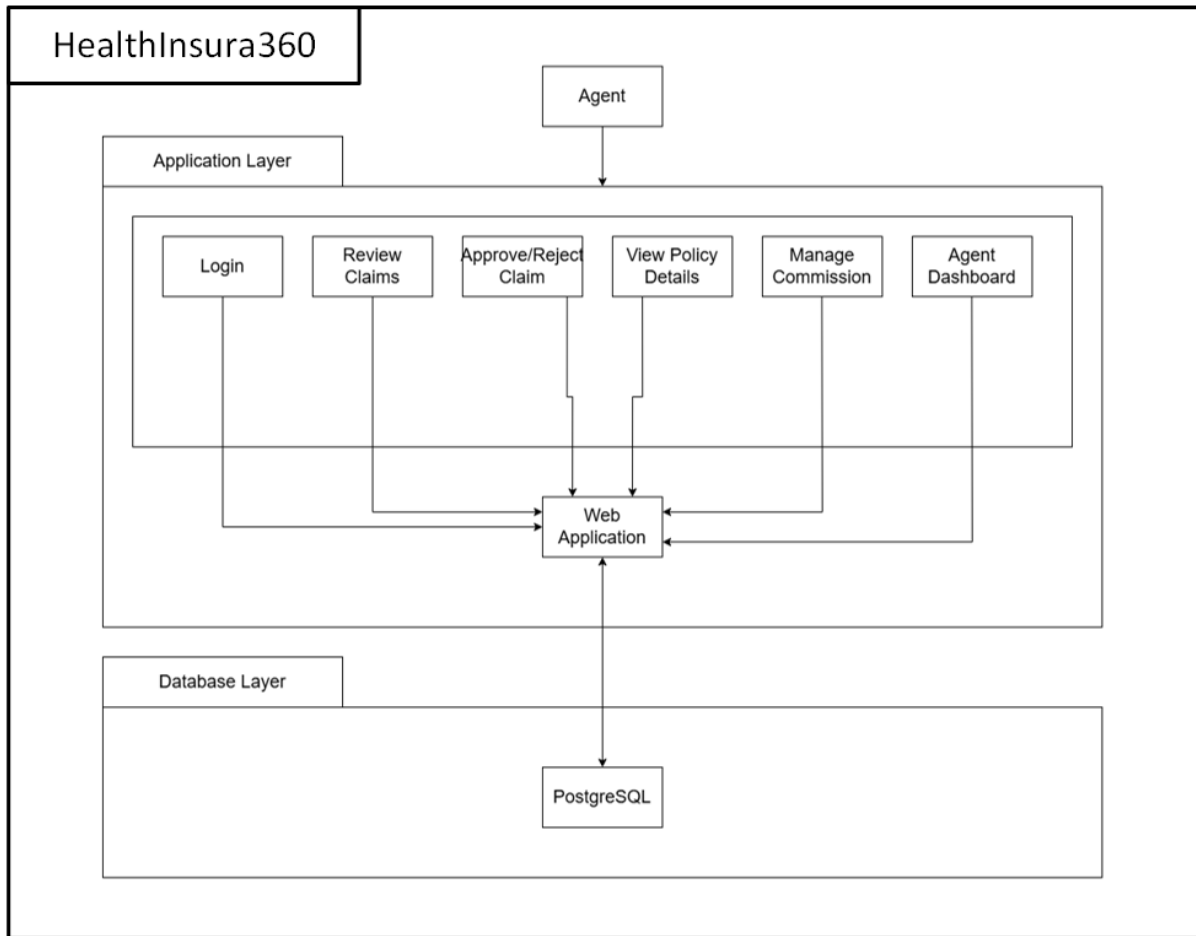


Figure 3. 4: Agent Package Diagram

3.2.2 Deployment Diagram (Physical)

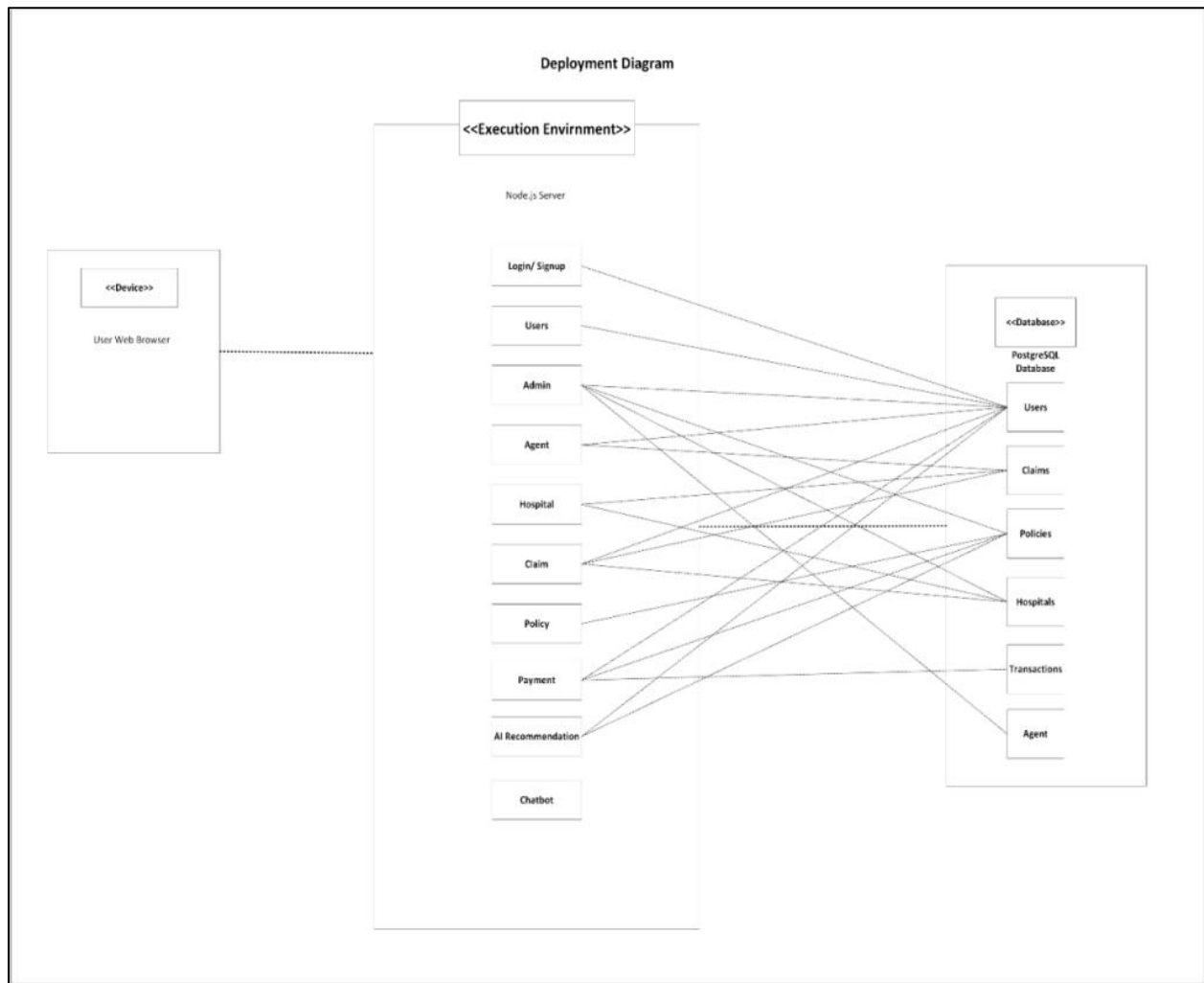


Figure 3. 5: Deployment Diagram

3.3 Logical Viewpoint

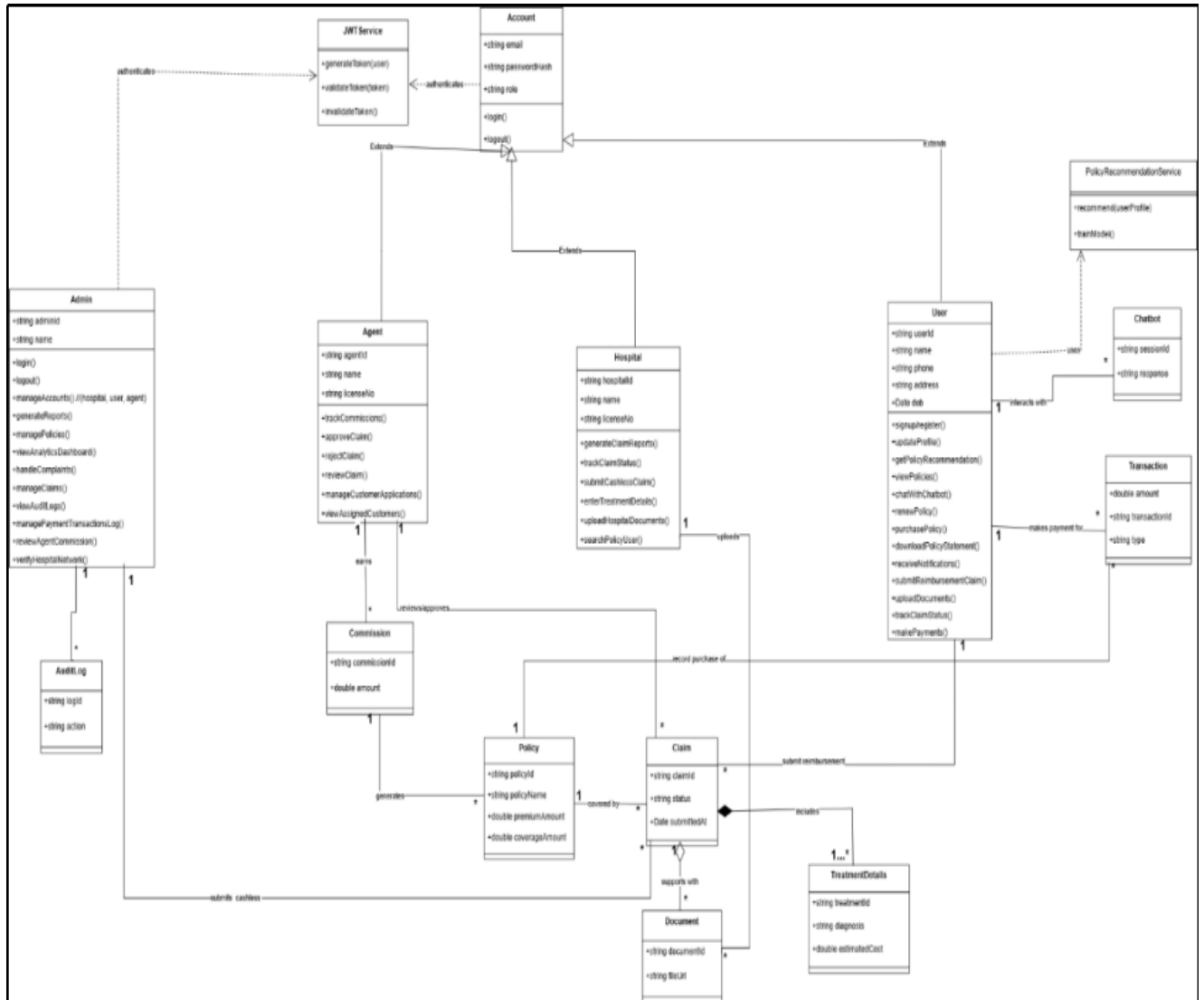


Figure 3. 6: Class Diagram

3.4 Information Viewpoint

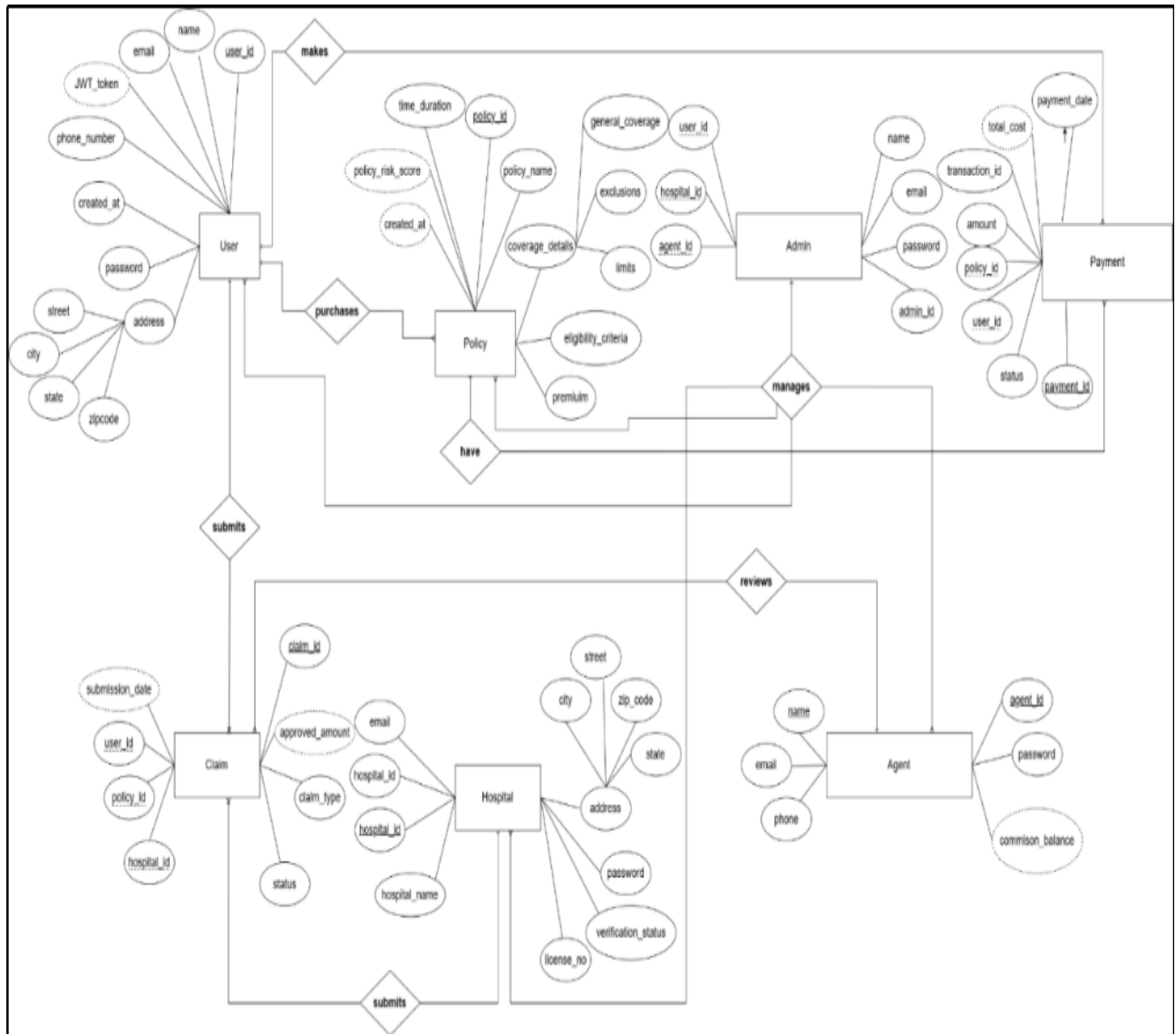


Figure 3. 7: Entity Relationship Diagram

3.5 Interaction Viewpoint

3.5.1 Sequence Diagram for User

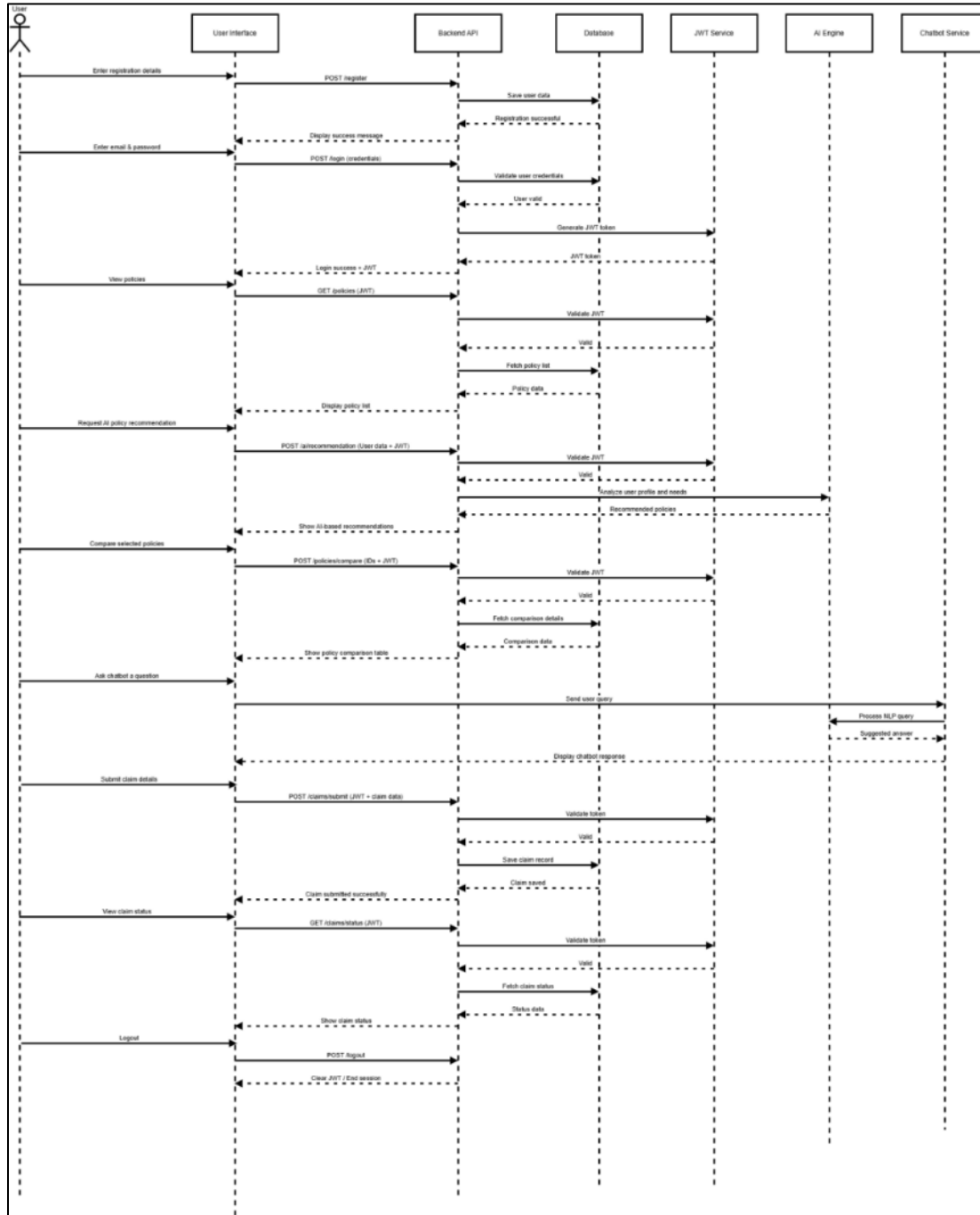


Figure 3. 8: Sequence Diagram of User

3.5.2 Sequence Diagram for Admin

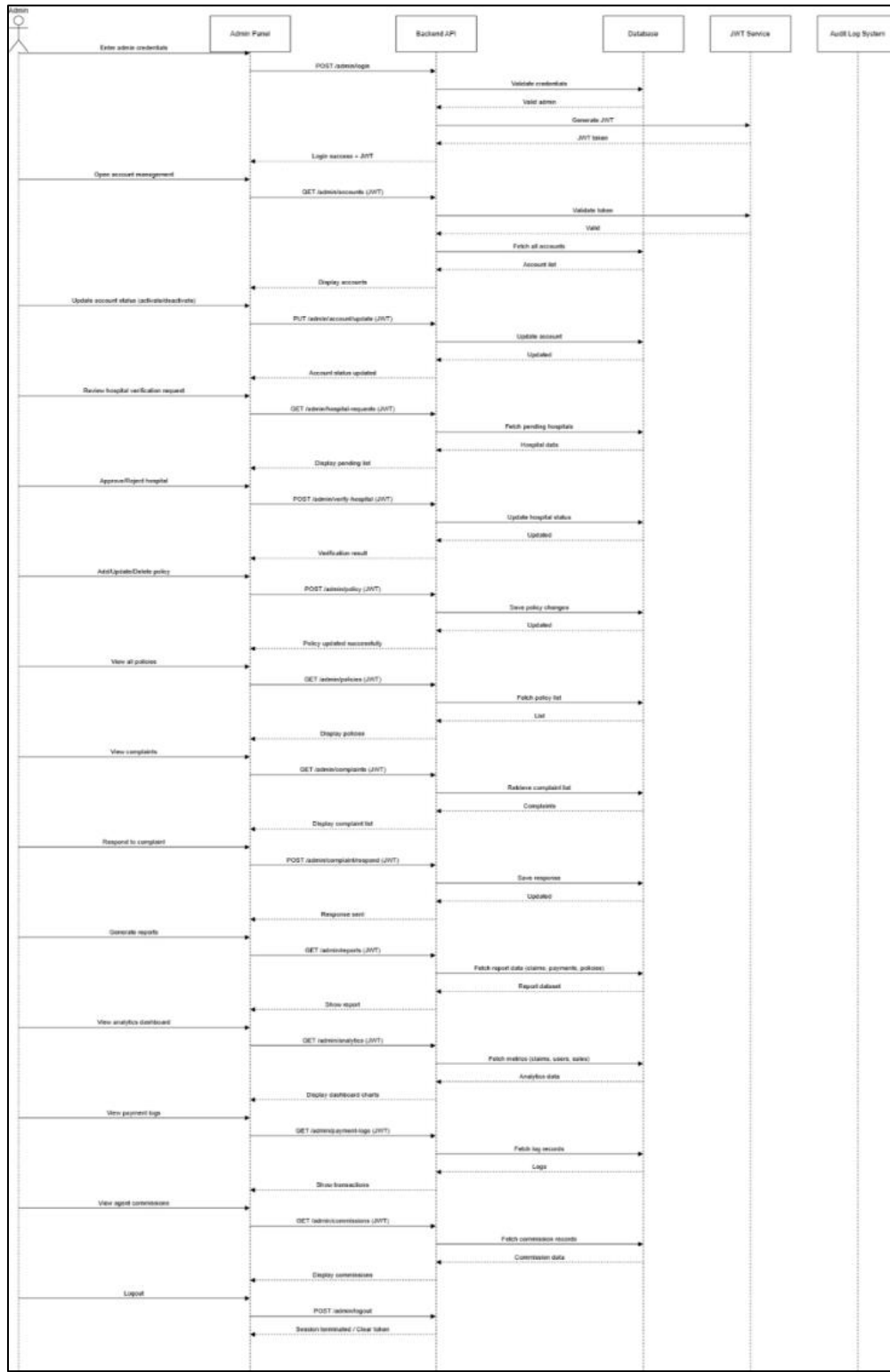


Figure 3. 9: Sequence Diagram of Admin

3.5.3 Sequence Diagram for Agent

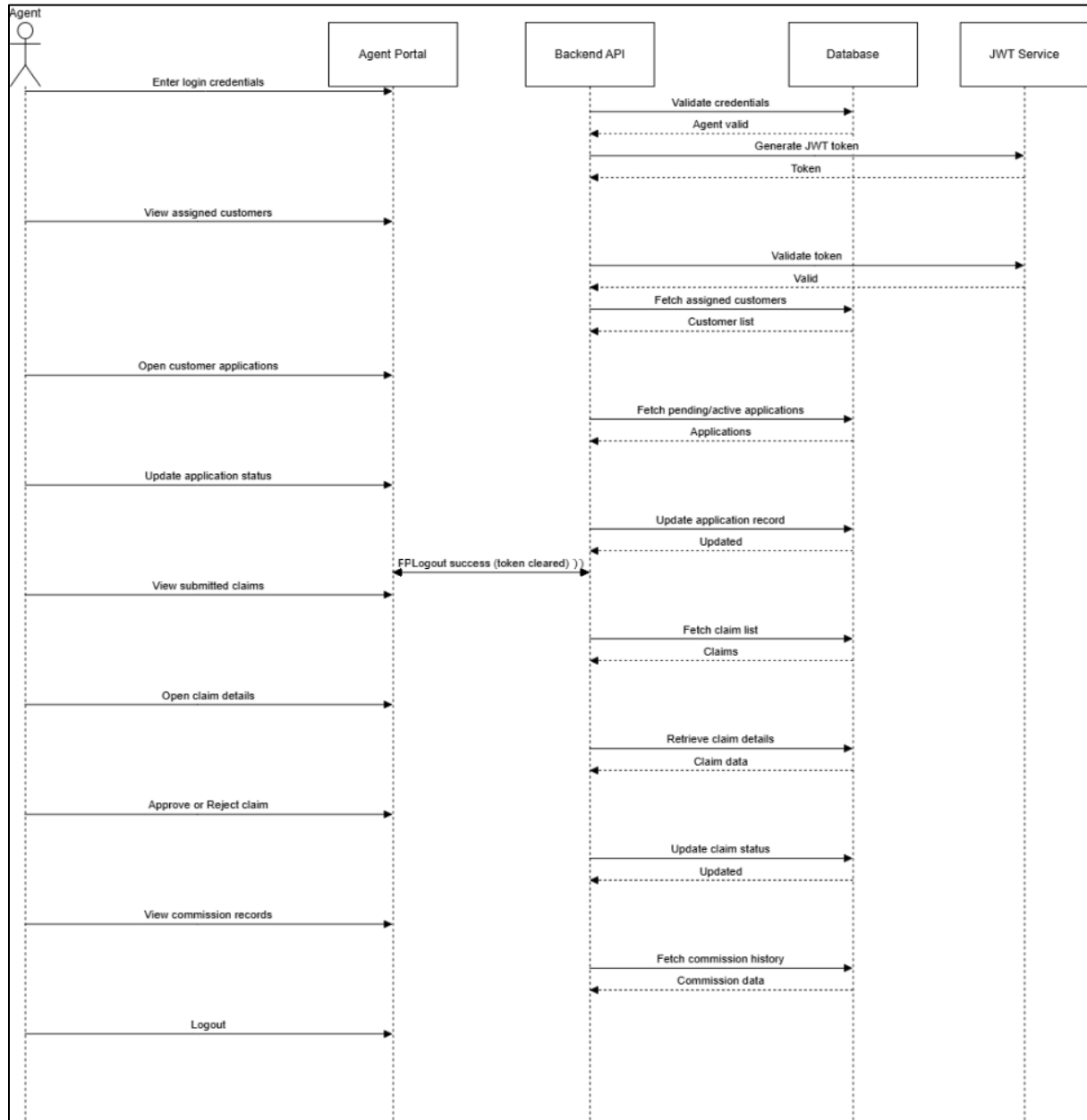


Figure 3. 10: Sequence Diagram of Agent

3.5.4 Sequence Diagram for Hospital

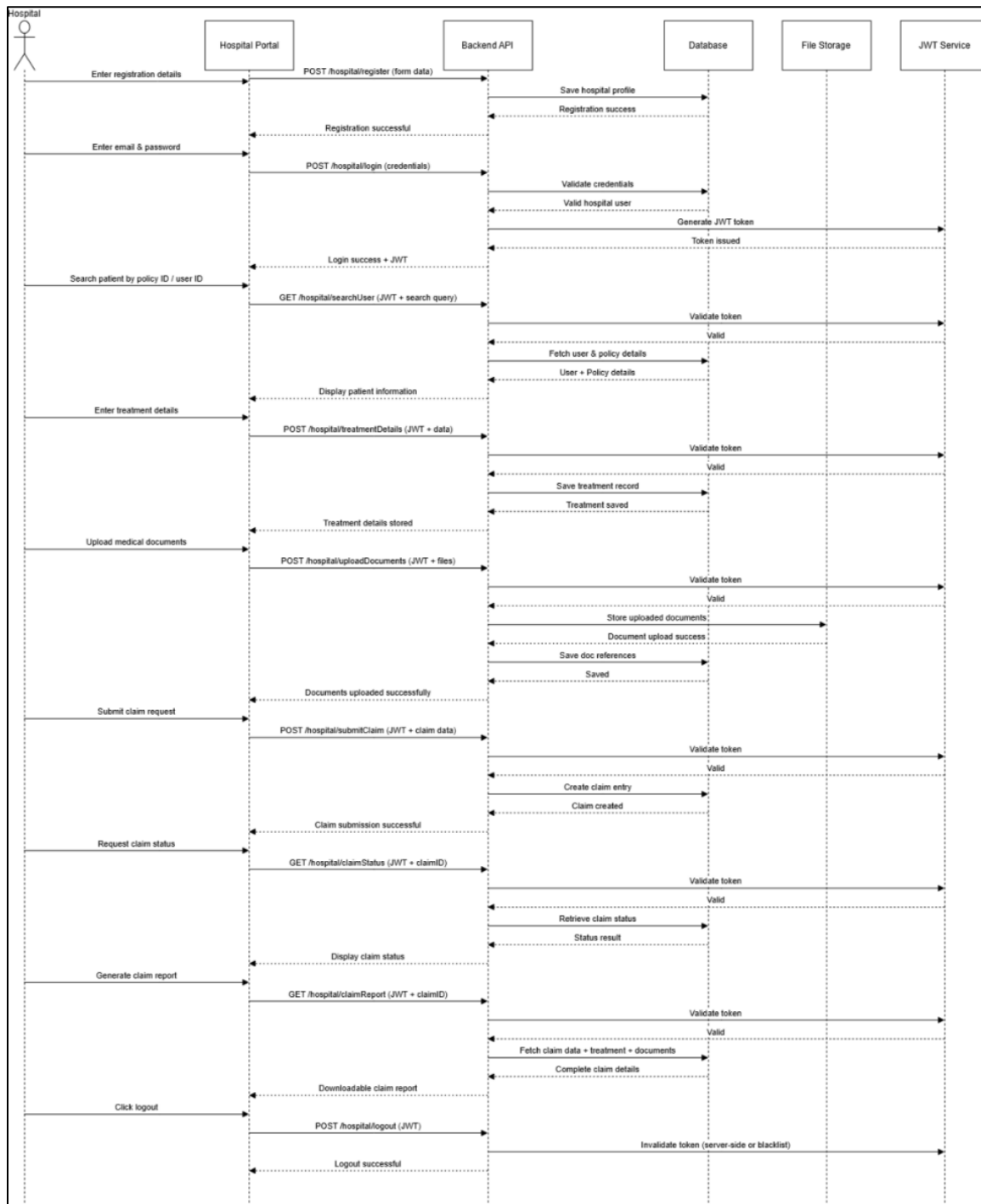


Figure 3. 11: Sequence Diagram of Hospital

3.6 State Dynamics Viewpoint

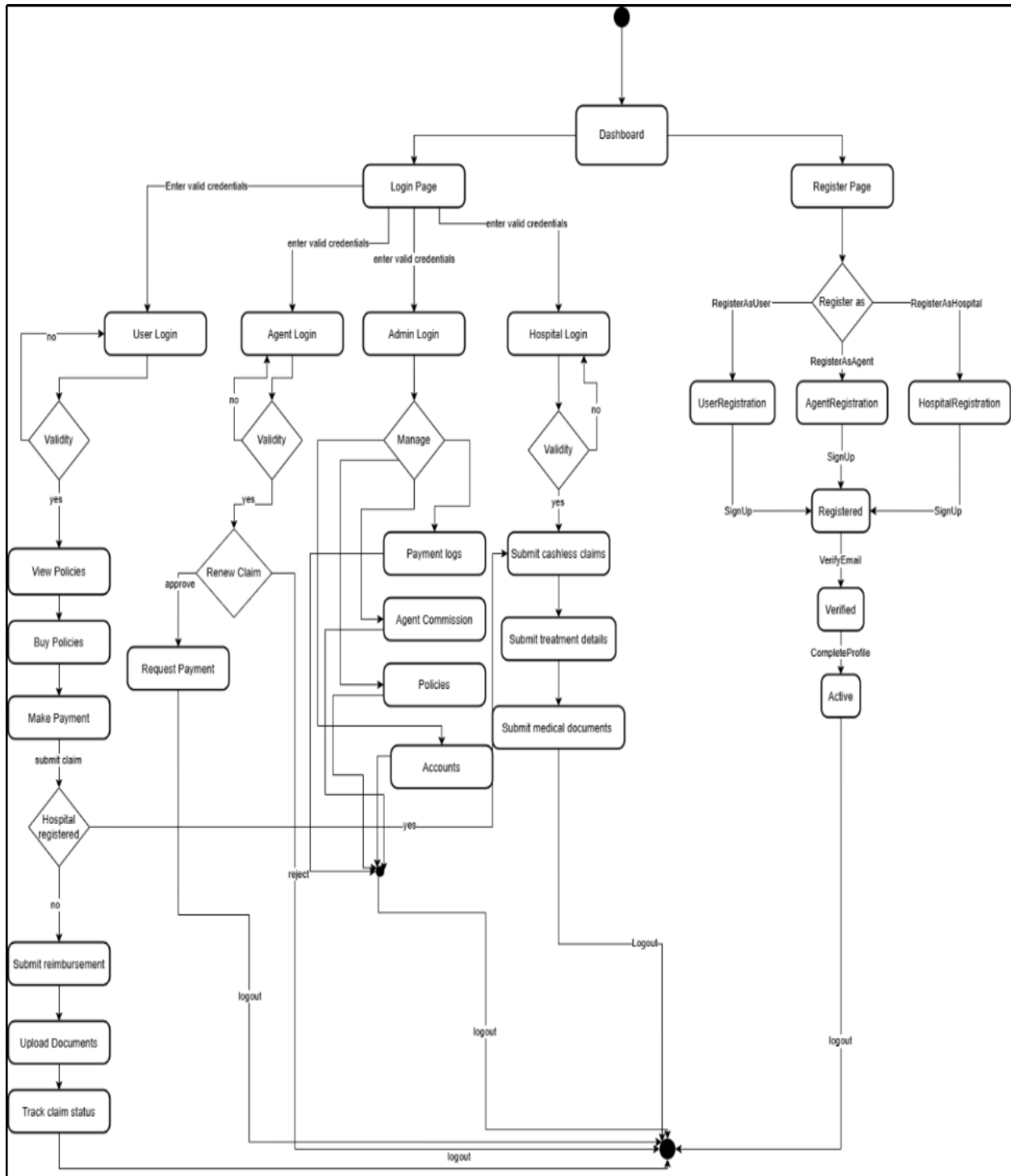


Figure 3. 12: State Machine Diagram

3.7 Algorithm Viewpoint

3.7.1 Pseudocode of the System

START SYSTEM

LOAD system configurations

CONNECT to database

INITIALIZE authentication service (JWT)

INITIALIZE file upload service

INITIALIZE AI recommendation engine

INITIALIZE notification service

INITIALIZE payment gateway

WHILE system is running:

 WAIT for any incoming request from (User, Admin, Agent, Hospital)

 IF request is "REGISTER":

 validate registration data

 IF email/phone exists:

 return error

 hash password

 store user/agent/hospital record

send welcome notification

continue

IF request is "LOGIN":

validate credentials

IF invalid:

return login failed

generate JWT token

generate refresh token

store active session

redirect to respective dashboard (user/admin/agent/hospital)

continue

IF request contains JWT token:

verify token

IF expired/invalid:

request re-authentication

USER OPERATIONS

IF actor == USER:

IF request == "UPDATE PROFILE":

validate input

update user profile

return success

IF request == "VIEW POLICIES":

fetch all active policies

return policy list

IF request == "AI POLICY RECOMMENDATION":

fetch user data

build recommendation feature vector

recommendedPolicies = AI.generate()

IF none:

recommendedPolicies = ruleBasedSuggestions()

return recommendedPolicies

IF request == "COMPARE POLICIES":

fetch selected policy details

return comparison result

IF request == "PURCHASE POLICY":

calculate premium

initiate payment process

IF payment success:

activate policy

issue policy document

notify user

ELSE:

mark payment failed

continue

IF request == "RENEW POLICY":

calculate renewal premium

initiate payment

IF payment success:

extend policy validity

notify renewal success

continue

IF request == "DOWNLOAD POLICY STATEMENT":

verify policy ownership

retrieve PDF

send to user

IF request == "SUBMIT REIMBURSEMENT CLAIM":

validate policy

create claim record

upload documents

assign claim to agent

notify user

continue

IF request == "UPLOAD CLAIM DOCUMENT":

 validate file type

 store in cloud

 link to claim

 continue

IF request == "TRACK CLAIM STATUS":

 retrieve claim progress

 return status

IF request == "CHAT WITH CHATBOT":

 process user input

 respond using AI chatbot engine

IF request == "VIEW NOTIFICATIONS":

 fetch all notifications

 return list

IF request == "MAKE PAYMENT":

 initiate payment gateway

 update payment transaction log

 return success or failure

IF request == "LOGOUT":

invalidate session tokens

return success

ADMIN OPERATIONS

IF actor == ADMIN:

IF request == "MANAGE ACCOUNTS":

fetch account list (user/agent/hospital)

perform (activate/deactivate/delete/update)

log activity

continue

IF request == "MANAGE POLICIES":

create/update/delete policy

return success

IF request == "GENERATE REPORTS":

compile dataset (claims, finance, policies)

format as PDF/CSV

return generated report

IF request == "VIEW ANALYTICS DASHBOARD":

aggregate system metrics

return analytics graphs & KPIs

IF request == "HANDLE COMPLAINTS":

fetch complaint list

update status after resolution

notify user

IF request == "MANAGE CLAIMS":

fetch all claims

approve/reject/escalate

notify respective actor

IF request == "VIEW AUDIT LOGS":

return logs sorted by timestamp

IF request == "MANAGE PAYMENT TRANSACTION LOG":

fetch all payment transactions

filter by date/type/etc.

IF request == "REVIEW AGENT COMMISSION":

calculate commission totals

return summary

IF request == "VERIFY HOSPITAL NETWORK":

check hospital documents

approve/reject/request more documents

notify hospital

IF request == "LOGOUT":

invalidate token

return success

AGENT OPERATIONS

IF actor == AGENT:

IF request == "VIEW ASSIGNED CUSTOMERS":

fetch list of mapped customers

return list

IF request == "MANAGE APPLICATIONS":

approve/reject new policy applications

notify users

IF request == "REVIEW CLAIM":

load claim details

check submitted documents

return claim analysis

IF request == "APPROVE CLAIM":

update claim status APPROVED

notify user & hospital

IF request == "REJECT CLAIM":

update claim status REJECTED

notify user & hospital

IF request == "TRACK COMMISSION":

calculate sum of earned commissions

return commission summary

IF request == "LOGOUT":

invalidate tokens

HOSPITAL OPERATIONS

IF actor == HOSPITAL:

IF request == "SEARCH USER POLICY":

search user by policy number or ID

return policy details

IF request == "ENTER TREATMENT DETAILS":

save treatment summary

attach to claim

IF request == "UPLOAD HOSPITAL DOCUMENTS":

 validate & upload files

 link to claim

IF request == "SUBMIT CASHLESS CLAIM":

 validate user policy

 create cashless claim entry

 assign agent

 notify user

IF request == "GENERATE CLAIM REPORT":

 fetch all hospital claims

 generate PDF summary

 return file

IF request == "TRACK CLAIM STATUS":

 retrieve claim progress

 return status

IF request == "LOGOUT":

 invalidate token

END IF

END WHILE

SHUTDOWN SERVICES

END SYSTEM

4: DEVELOPMENT AND TOOLS

4.1 Introduction

This chapter explains the development procedure, team contributions, tools used, and future extensions.

4.2 Development Plan

Table 4. 1: Development Plan

Task Name	Duration	Timeline	Status
Project Planning	5 days	01/09/25 - 05/09/25	100%
Proposal	10 days	08/09/25 - 18/09/25	100%
Requirements Gathering	2 weeks	22/09/25 - 05/10/25	100%
SRDS Documentation	1 month	06/10/25 - 06/11/25	100%
Design Phase (Diagrams)	2 weeks	10/11/25 - 24/11/25	100%
Database Design	20 days	25/11/25 - 15/12/25	100%
Backend Development	1.5 month	16/12/25 - 30/01/26	100%
Frontend Development	1 month	06/02/26 - 06/03/26	100%
AI Integration	2 weeks	07/03/26 - 20/03/26	100%
Testing	2 weeks	23/03/26 - 06/04/26	100%
Final Report	2 weeks	12/04/26 - 25/04/26	100%

4.3 Development Tools

- **Payment Gateway:** Stripe API
- **Design:** Figma
- **Diagrams:** [Draw.io](https://draw.io), Lucid chart
- **IDE:** Visual Studio Code
- **Database:** PostgreSQL
- **Frontend:** React.js
- **Backend:** Node.js, Express.js

- **Version Control:** Git, GitHub
- **API Testing:** Postman

4.4 Conclusion and Future Work

Conclusion:

HealthInsura360 successfully delivers an AI-powered health insurance management platform that addresses the key pain points in traditional insurance systems. The integration of AI recommendations, chatbot assistance, digital claim processing, and secure payments creates a comprehensive solution for all stakeholders.

Future Extensions:

- i. Mobile application development (iOS and Android)
- ii. Integration with real banking APIs
- iii. Blockchain-based claim verification
- iv. Advanced predictive analytics for fraud detection
- v. Telemedicine integration
- vi. Multi-language support
- vii. Voice-enabled chatbot
- viii. Integration with government health databases

5: QUALITY ASSURANCE

5.1 Introduction

This chapter focuses on the quality assurance process, including traceability matrix, test plan, and test cases to ensure HealthInsura360 is reliable, accurate, and provides an exceptional user experience.

5.2 Traceability Matrix

Table 4. 2: Traceability Matrix

	Test Case ID	TC-1	TC-2	TC-3	TC-4	TC-5	TC-6	TC-7	TC-8	TC-9	TC-10	TC-11	TC-12	TC-13	TC-14	TC-15	TC-16	TC-17	TC-18	TC-19	TC-20	TC-21	TC-22	TC-23	TC-24	Test Cases For Respective Requirements
Requirement Name																										
User Registration		✖																								1
User Login			✖																							1
View Policies				✖																						1
AI Recommendations					✖																					1
Chat with Chatbot						✖																				1
Purchase Policy							✖																			1
Submit Claim								✖																		1
Track Claim Status									✖																	1
Make Payment							✖			✖																2
Agent Login											✖															1

[illegible]

5.3 Test Plan

5.3.1 Test Cases for User Module

Test Case TC-01: User Registration

Table 5. 1: Test Case 1

Test ID	TC-01
Test Name	User Registration with OTP Verification
Date Of Test	23/03/26
Name Of Application	HealthInsura360
Description	Verify user can create account with OTP verification
Pre-condition	User must not have existing account, valid email/phone
Input	Name, email, phone number, password
Expected Output	Account created successfully, OTP verified, redirect to login
Actual Output	As expected
Test Role (Actor)	User
Test Verified By	Team Members

Test Case TC-02: User Login

Table 5. 2: Test Case 2

Test ID	TC-02
Test Name	User Login with JWT Authentication
Date Of Test	23/03/26
Name Of Application	HealthInsura360
Description	Verify user can securely login with valid credentials
Pre-condition	User must have registered account
Input	Email/phone, password
Expected Output	JWT token generated, user redirected to dashboard
Actual Output	As expected
Test Role (Actor)	User
Test Verified By	Team Members

Test Case TC-03: User View Policies

Table 5. 3: Test Case 3

Test ID	TC-03
Test Name	View and Filter Available Policies
Date Of Test	23/03/26
Name Of Application	HealthInsura360
Description	Verify user can browse and filter insurance policies
Pre-condition	User must be logged in, policies available in database
Input	Filter criteria (category, price, coverage)
Expected Output	List of policies displayed matching filter criteria
Actual Output	As expected
Test Role (Actor)	User
Test Verified By	Team Members

Test Case TC-04: User Get AI Policy Recommendations

Table 5. 4: Test Case 4

Test ID	TC-04
Test Name	AI-Based Policy Recommendations
Date Of Test	23/03/26
Name Of Application	HealthInsura360

Description	Verify system suggests personalized policies using AI
Pre-condition	User must be logged in with complete profile
Input	User views dashboard, AI recommendations are displayed automatically
Expected Output	Personalized policy suggestions displayed
Actual Output	As expected
Test Role (Actor)	User
Test Verified By	Team Members

Test Case TC-05: User Chat with Chatbot

Table 5. 5: Test Case 5

Test ID	TC-05
Test Name	AI Chatbot Interaction
Date Of Test	24/03/26
Name Of Application	HealthInsura360
Description	Verify user can chat with AI chatbot for queries
Pre-condition	User must be logged in
Input	User types a question about policies
Expected Output	Chatbot responds with relevant information
Actual Output	As expected
Test Role (Actor)	User
Test Verified By	Team Members

Test Case TC-06: User Purchase Policy

Table 5. 6: Test Case 6

Test ID	TC-06
Test Name	Policy Purchase with Payment
Date Of Test	24/03/26
Name Of Application	HealthInsura360
Description	Verify user can purchase policy and make payment
Pre-condition	User logged in, policy selected, payment method available
Input	Select policy, enter payment details
Expected Output	Payment successful, policy activated, document generated
Actual Output	As expected
Test Role (Actor)	User
Test Verified By	Team Members

Test Case TC-07: User Submit Claim

Table 5. 7: Test Case 7

Test ID	TC-07
Test Name	Submit Reimbursement Claim
Date Of Test	24/03/26
Name Of Application	HealthInsura360
Description	Verify user can submit claim with medical documents
Pre-condition	User must have active policy
Input	Treatment details, uploaded medical bills/documents
Expected Output	Claim submitted successfully, assigned to agent
Actual Output	As expected
Test Role (Actor)	User
Test Verified By	Team Members

Test Case TC-08: User Track Claim Status

Table 5. 8: Test Case 8

Test ID	TC-08
Test Name	Track Claim Status
Date Of Test	24/03/26
Name Of Application	HealthInsura360
Description	Verify user can track progress of submitted claims
Pre-condition	Claim must exist in system
Input	User navigates to "Track Claim Status"
Expected Output	Current claim status displayed (Pending/Approved/Rejected)
Actual Output	As expected
Test Role (Actor)	User
Test Verified By	Team Members

Test Case TC-09: User Make Payment

Table 5. 9: Test Case 9

Test ID	TC-09
Test Name	Secure Payment Processing

Date Of Test	26/03/26
Name Of Application	HealthInsura360
Description	Verify user can make payment via payment gateway
Pre-condition	User logged in, policy selected for purchase/renewal
Input	Payment method, card details
Expected Output	Payment processed, receipt generated
Actual Output	As expected
Test Role (Actor)	User
Test Verified By	Team Members

5.3.2 Test Cases for Agent Module

Test Case TC-10: Agent Login

Table 5. 10: Test Case 10

Test ID	TC-10
Test Name	Agent Login with JWT Authentication
Date Of Test	26/03/26
Name Of Application	HealthInsura360
Description	Verify agent can securely login to agent portal
Pre-condition	Agent account must be created and approved
Input	Agent ID/email, password
Expected Output	JWT token generated, agent dashboard displayed
Actual Output	As expected
Test Role (Actor)	Agent
Test Verified By	Team Members

Test Case TC-11: Agent View Assigned Customers

Table 5. 11: Test Case 11

Test ID	TC-11
Test Name	View Assigned Customers
Date Of Test	26/03/26
Name Of Application	HealthInsura360
Description	Verify agent can view list of assigned customers
Pre-condition	Agent must be logged in, customers assigned

Input	Agent navigates to "Assigned Customers"
Expected Output	List of assigned customers displayed
Actual Output	As expected
Test Role (Actor)	Agent
Test Verified By	Team Members

Test Case TC-12: Agent Review and Approve Claim

Table 5. 12: Test Case 12

Test ID	TC-12
Test Name	Review and Approve Claim
Date Of Test	26/03/26
Name Of Application	HealthInsura360
Description	Verify agent can review documents and approve claim
Pre-condition	Claim submitted by user, assigned to agent
Input	Agent reviews documents, clicks "Approve"
Expected Output	Claim status updated to "Approved", user notified
Actual Output	As expected
Test Role (Actor)	Agent
Test Verified By	Team Members

Test Case TC-13: Agent Reject Claim

Table 5. 13: Test Case 13

Test ID	TC-13
Test Name	Review and Reject Claim
Date Of Test	27/03/26
Name Of Application	HealthInsura360
Description	Verify agent can reject invalid claims with reason
Pre-condition	Claim submitted by user, assigned to agent
Input	Agent reviews documents, clicks "Reject", adds reason
Expected Output	Claim status updated to "Rejected", user notified
Actual Output	As expected
Test Role (Actor)	Agent
Test Verified By	Team Members

Test Case TC-14: Agent Track Commissions

Table 5. 14: Test Case 14

Test ID	TC-14
Test Name	Track Agent Commissions
Date Of Test	27/03/26
Name Of Application	HealthInsura360
Description	Verify agent can track earned commissions
Pre-condition	Agent must have policy sales recorded
Input	Agent navigates to "Commission Dashboard"
Expected Output	Commission summary displayed
Actual Output	As expected
Test Role (Actor)	Agent
Test Verified By	Team Members

5.3.3 Test Cases for Hospital Module

Test Case TC-15: Hospital Login

Table 5. 15: Test Case 15

Test ID	TC-15
Test Name	Hospital Login
Date Of Test	28/03/26
Name Of Application	HealthInsura360
Description	Verify hospital can login to hospital portal
Pre-condition	Hospital account must be registered and approved
Input	Hospital ID/email, password
Expected Output	Hospital dashboard displayed
Actual Output	As expected
Test Role (Actor)	Hospital
Test Verified By	Team Members

Test Case TC-16: Hospital Search Patient Policy

Table 5. 16: Test Case 16

Test ID	TC-16
Test Name	Search and Verify Patient Policy

Date Of Test	28/03/26
Name Of Application	HealthInsura360
Description	Verify hospital can search patient's policy details
Pre-condition	Hospital logged in, patient has valid policy
Input	Policy number or patient ID
Expected Output	Patient policy details displayed
Actual Output	As expected
Test Role (Actor)	Hospital
Test Verified By	Team Members

Test Case TC-17: Hospital Submit Cashless Claim

Table 5. 17: Test Case 17

Test ID	TC-17
Test Name	Submit Cashless Claim
Date Of Test	29/03/26
Name Of Application	HealthInsura360
Description	Verify hospital can submit cashless claim for patient
Pre-condition	Patient policy verified, patient admitted
Input	Treatment details, medical reports, estimated amount
Expected Output	Cashless claim submitted, forwarded to agent
Actual Output	As expected
Test Role (Actor)	Hospital
Test Verified By	Team Members

Test Case TC-18: Hospital Upload Documents and Track Status

Table 5. 18: Test Case 18

Test ID	TC-18
Test Name	Upload Medical Documents and Track Claim Status
Date Of Test	29/03/26
Name Of Application	HealthInsura360
Description	Verify hospital can upload documents and track claim
Pre-condition	Claim submitted by hospital
Input	Upload medical files, click "Track Status"
Expected Output	Documents uploaded successfully, claim status displayed
Actual Output	As expected
Test Role (Actor)	Hospital
Test Verified By	Team Members

5.3.4 Test Cases for Admin Module

Test Case TC-19: Admin Login

Table 5. 19: Test Case 19

Test ID	TC-19
Test Name	Admin Login
Date Of Test	
Name Of Application	HealthInsura360
Description	Verify admin can login to admin panel
Pre-condition	Admin account must be created
Input	Admin email, password
Expected Output	Admin dashboard displayed
Actual Output	As expected
Test Role (Actor)	Admin
Test Verified By	Team Members

Test Case TC-20: Admin Manage User Accounts

Table 5. 20: Test Case 20

Test ID	TC-20
Test Name	Manage User Accounts (CRUD)
Date Of Test	01/04/26
Name Of Application	HealthInsura360
Description	Verify admin can add, update, delete user accounts
Pre-condition	Admin logged in
Input	Add/Edit/Delete user details
Expected Output	User account updated successfully in database
Actual Output	As expected
Test Role (Actor)	Admin
Test Verified By	Team Members

Test Case TC-21: Admin Manage Policies

Table 5. 21: Test Case 21

Test ID	TC-21
Test Name	Manage Policies (Add/Edit/Delete)

Date Of Test	02/04/26
Name Of Application	HealthInsura360
Description	Verify admin can create, update, delete policies
Pre-condition	Admin logged in
Input	Policy details (name, premium, coverage, etc.)
Expected Output	Policy added/updated/deleted successfully
Actual Output	As expected
Test Role (Actor)	Admin
Test Verified By	Team Members

Test Case TC-22: Admin Generate Reports

Table 5. 22: Test Case 22

Test ID	TC-22
Test Name	Generate System Reports
Date Of Test	03/04/26
Name Of Application	HealthInsura360
Description	Verify admin can generate claims and payment reports
Pre-condition	Admin logged in, data exists in system
Input	Select report type (Claims/Payments/Policies), date range
Expected Output	Report generated in PDF/CSV format
Actual Output	As expected
Test Role (Actor)	Admin
Test Verified By	Team Members

Test Case TC-23: Admin View Analytics Dashboard

Table 5. 23: Test Case 23

Test ID	TC-23
Test Name	View Analytics Dashboard
Date Of Test	03/04/26
Name Of Application	HealthInsura360
Description	Verify admin can view system analytics and metrics
Pre-condition	Admin logged in, system has data
Input	Admin navigates to Dashboard
Expected Output	Analytics graphs and KPIs displayed
Actual Output	As expected
Test Role (Actor)	Admin
Test Verified By	Team Members

Test Case TC-24: Admin View Audit Logs

Table 5. 24: Test Case 24

Test ID	TC-24
Test Name	View System Audit Logs
Date Of Test	06/04/26
Name Of Application	HealthInsura360
Description	Verify admin can view audit logs of system activities
Pre-condition	Admin logged in with audit privileges
Input	Admin navigates to "Audit Logs"
Expected Output	List of system activities displayed with timestamps
Actual Output	As expected
Test Role (Actor)	Admin
Test Verified By	Team Members

6: USER MANUAL

6.1 Introduction

This user manual provides comprehensive instructions for using HealthInsura360. It covers system requirements, installation, and operating procedures for all user types.

6.2 Hardware/Software Requirements

6.2.1 Hardware Requirements

- **Processor:** Intel Core i5 (6th Generation or higher)
- **RAM:** 8 GB (16 GB recommended)
- **Storage:** 256 GB SSD
- **Internet:** 2 Mbps (10+ Mbps recommended)

6.2.2 Software Requirements

- **Operating System:** Windows 7+, macOS, Linux, Android, iOS
- **Browser:** Chrome, Firefox, Safari, Edge (latest versions)
- **Internet:** Active connection

6.2.3 Installation Guide

Web Application:

1. Open a modern web browser
2. Navigate to the HealthInsura360 URL
3. No installation required

Access:

- User Portal: localhost:3001/user-dashboard
- Agent Portal: localhost:3001/agent-dashboard
- Hospital Portal: localhost:3001/hospital-dashboard
- Admin Portal: localhost:3001/admin-dashboard

6.3 Operating Manual

6.3.1 User (Policyholder) Operations

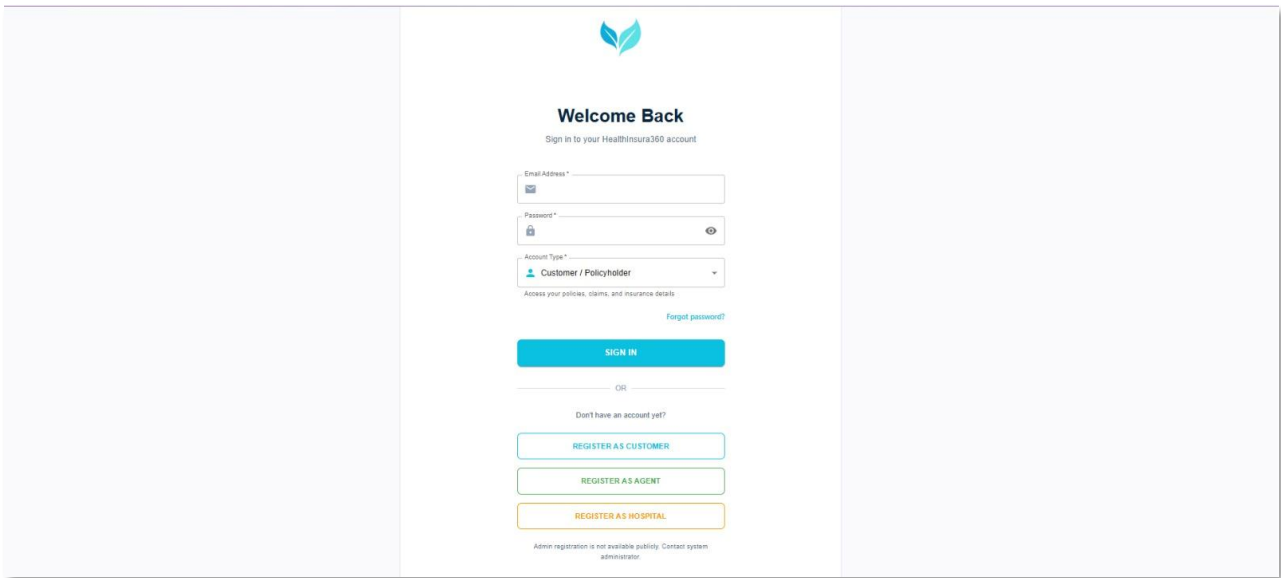
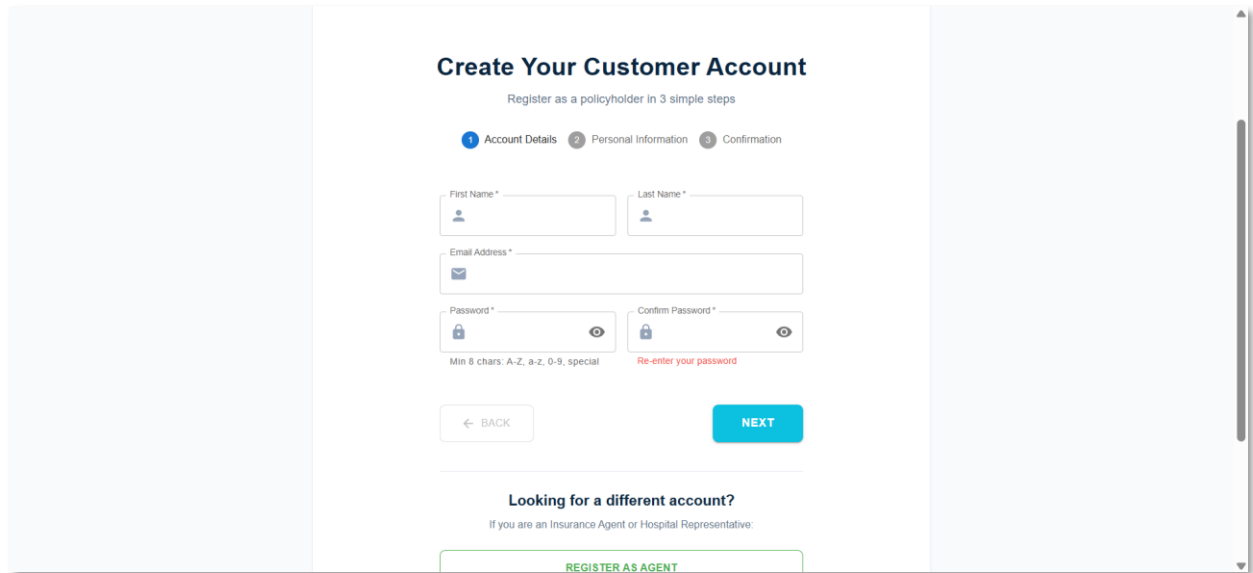
The image shows a user login screen for a system called HealthInsura360. At the top center is a logo consisting of two stylized leaves, one green and one blue. Below the logo, the text "Welcome Back" is displayed in bold, followed by "Sign in to your HealthInsura360 account" in a smaller font. The login form includes three input fields: "Email Address *" with an envelope icon, "Password *" with a lock icon and a toggle eye icon, and "Account Type *" with a dropdown menu currently showing "Customer / Policyholder". Below these fields is a link that says "Access your policies, claims, and insurance details" and another link that says "Forgot password?". A large blue button labeled "SIGN IN" is positioned below the links. Underneath the button is a horizontal line with the text "OR" in the center. Below this line, there are three buttons for registration: "REGISTER AS CUSTOMER" (blue), "REGISTER AS AGENT" (green), and "REGISTER AS HOSPITAL" (yellow). At the very bottom, a small note states: "Admin registration is not available publicly. Contact system administrator."

Figure 6. 1: User-Login Screen

Login:

1. Click "Login" on the homepage
2. Enter email/phone and password
3. Click "Sign In"
4. User dashboard loads



Create Your Customer Account
Register as a policyholder in 3 simple steps

1 Account Details 2 Personal Information 3 Confirmation

First Name * Last Name *

Email Address *

Password * Confirm Password *

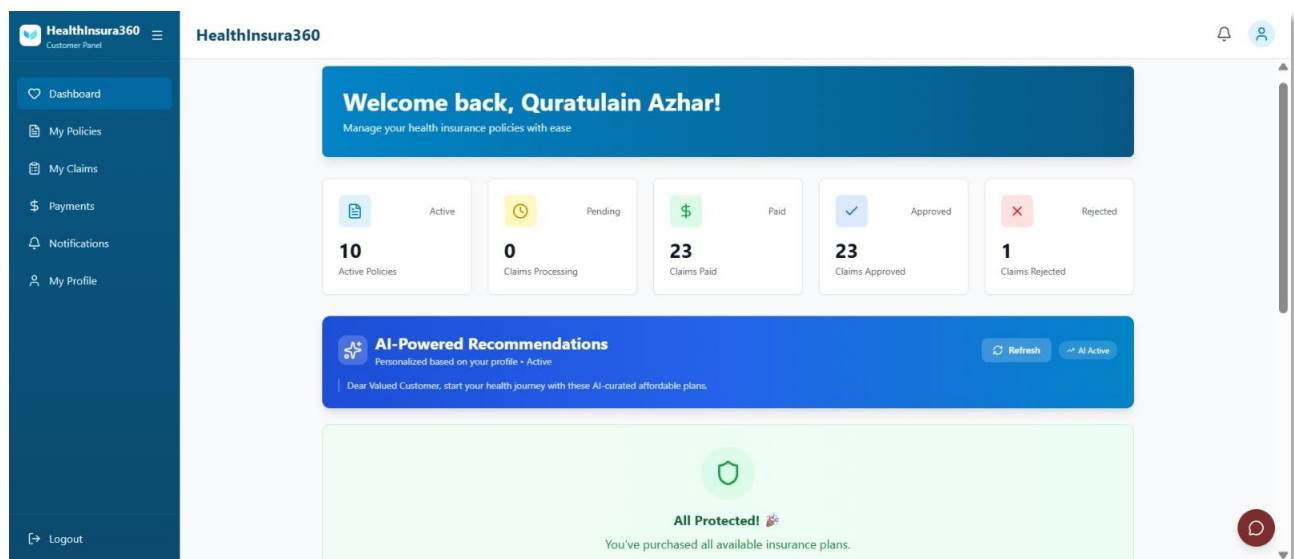
Min 8 chars: A-Z, a-z, 0-9, special Re-enter your password

[← BACK](#) [NEXT](#)

Looking for a different account?
If you are an Insurance Agent or Hospital Representative.

[REGISTER AS AGENT](#)

Figure 6. 2: User-Registration Screen



HealthInsura360 Customer Panel

HealthInsura360

Welcome back, Quratulain Azhar!
Manage your health insurance policies with ease

Active	Pending	Paid	Approved	Rejected
10 Active Policies	0 Claims Processing	23 Claims Paid	23 Claims Approved	1 Claims Rejected

AI-Powered Recommendations
Personalized based on your profile • Active

Dear Valued Customer, start your health journey with these AI-curated affordable plans.

[Refresh](#) [AI Active](#)

All Protected!
You've purchased all available insurance plans.

[Logout](#)

Figure 6. 3: User-Dashboard

Dashboard Overview:

After login, the user dashboard displays:

- Active policies summary
- Recent notifications

- Claim status
- AI policy recommendations

Get AI Recommendations:

1. On the dashboard, AI recommendations load automatically
2. Scroll through personalized policy suggestions
3. Click on any recommended policy to view details

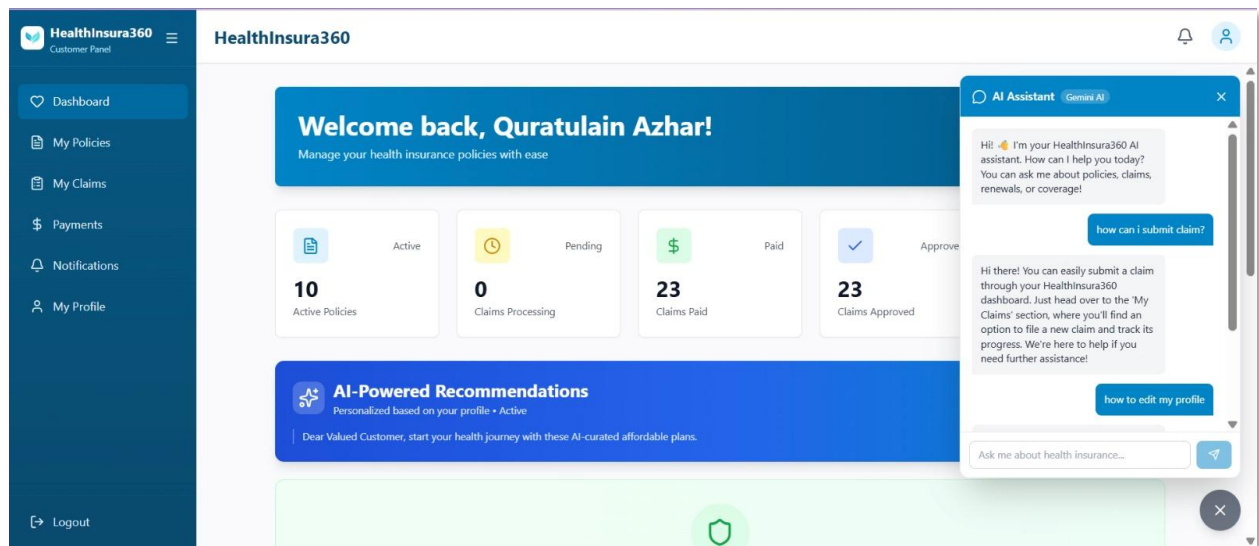


Figure 6. 4: User-Chat with Chatbot

Chat with Chatbot:

1. Click the orange chatbot icon at bottom right
2. Type your question (e.g., "How to submit a claim?")
3. Chatbot responds instantly with relevant information
4. Continue conversation for more help

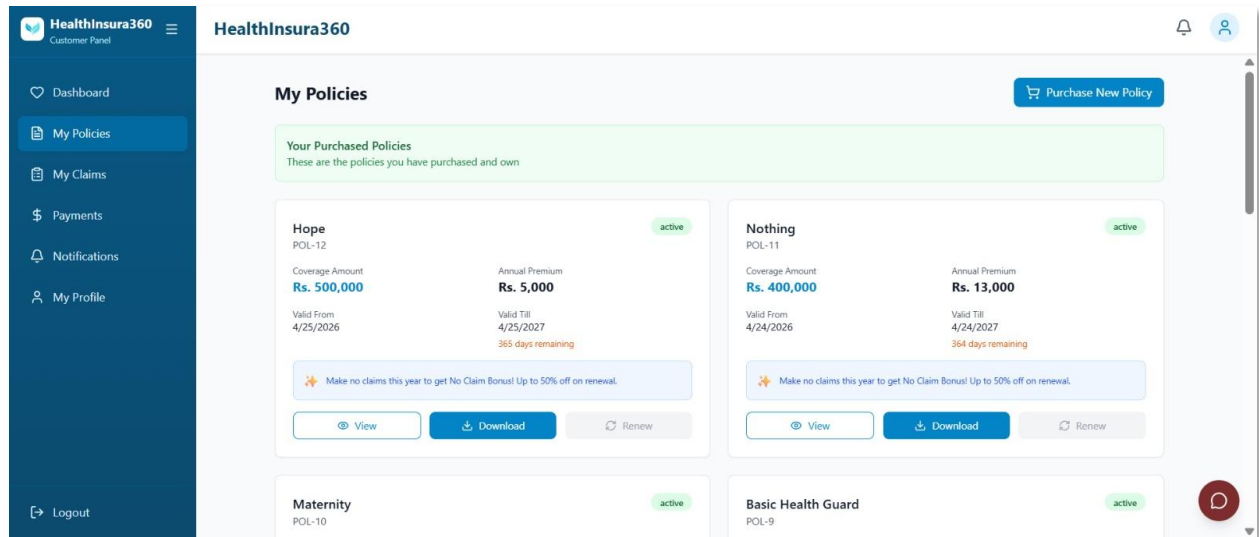


Figure 6. 5: User-View Policies

View Policies:

1. Click "Policies" in the navigation bar
2. Browse available health insurance plans
3. System automatically hides policies you already own
4. Click "View Details" on any policy for complete information

Purchase Policy:

1. After viewing policy details, click "Buy Now"
2. Review policy terms and premium amount
3. Click "Proceed to Payment"
4. Complete payment via Stripe
5. Policy is activated
6. Download policy document

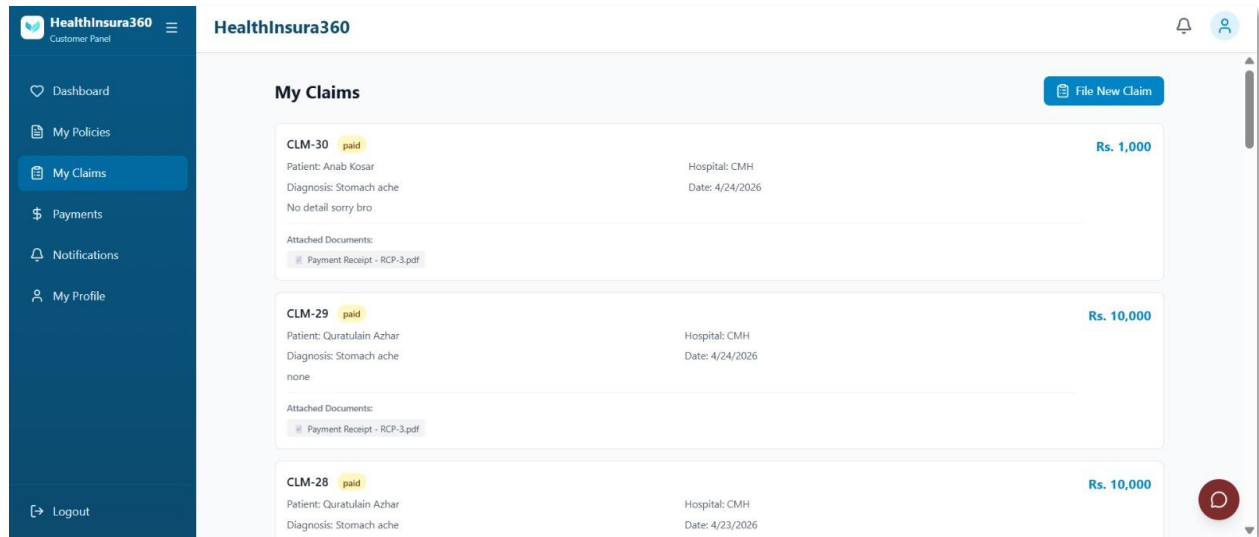


Figure 6. 6: User-Submit Claim

Submit Reimbursement Claim:

1. Navigate to "Claims" section
2. Click "Submit New Claim"
3. Select active policy
4. Fill treatment details (diagnosis, hospital name, dates)
5. Upload medical bills and documents
6. Click "Submit Claim"
7. Claim is assigned to agent for review

Track Claim Status:

1. Navigate to "Claims" section
2. View list of all submitted claims
3. Each claim shows status: Pending / Under Review / Approved / Rejected
4. Click on any claim to view detailed timeline

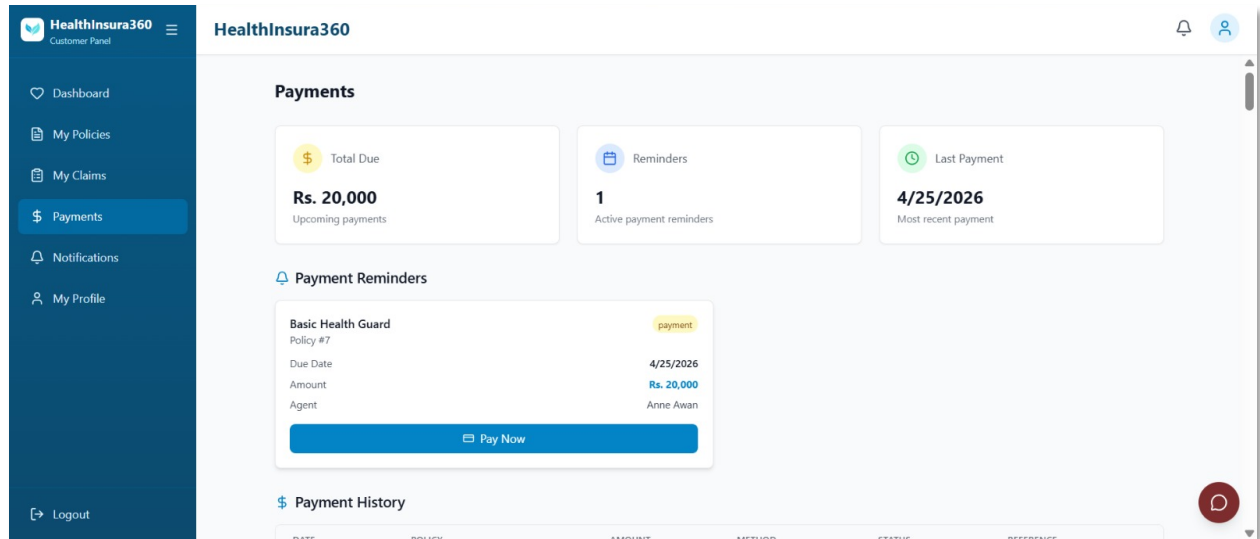


Figure 6. 7: User-Make Payments

Make Payment:

1. During policy purchase or renewal, go to payment page
2. Enter card details
3. Click "Pay Now"
4. Payment processed
5. Receipt generated and downloadable

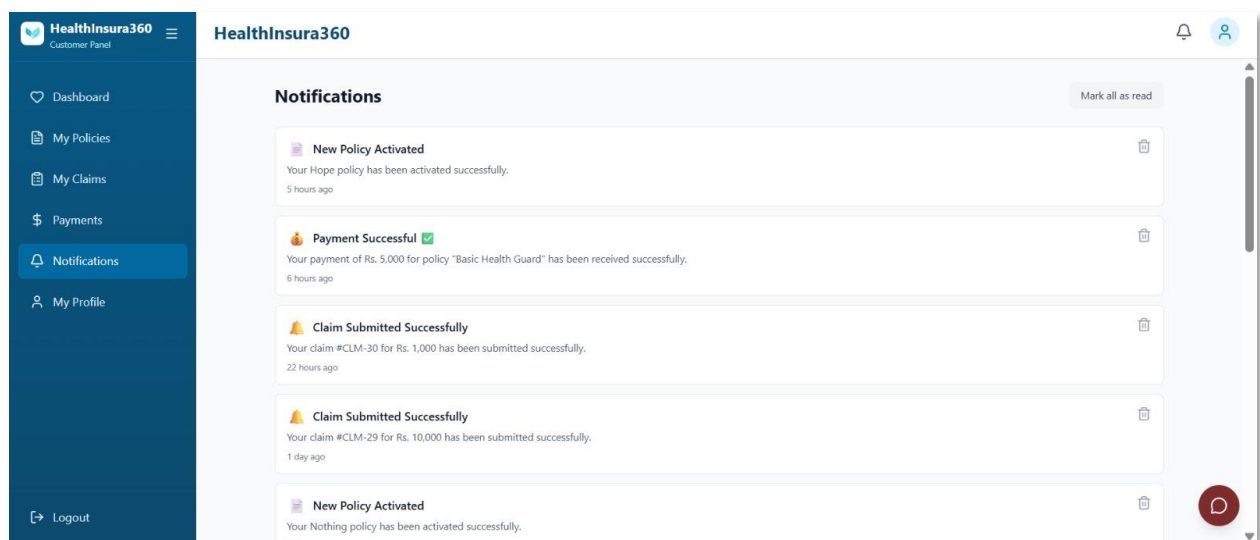


Figure 6. 8: User-View Notifications

View Notifications:

1. Click bell icon at top right
2. Dropdown shows recent notifications
3. Click "Mark as Read" to acknowledge
4. Click "View All" to see full notification history

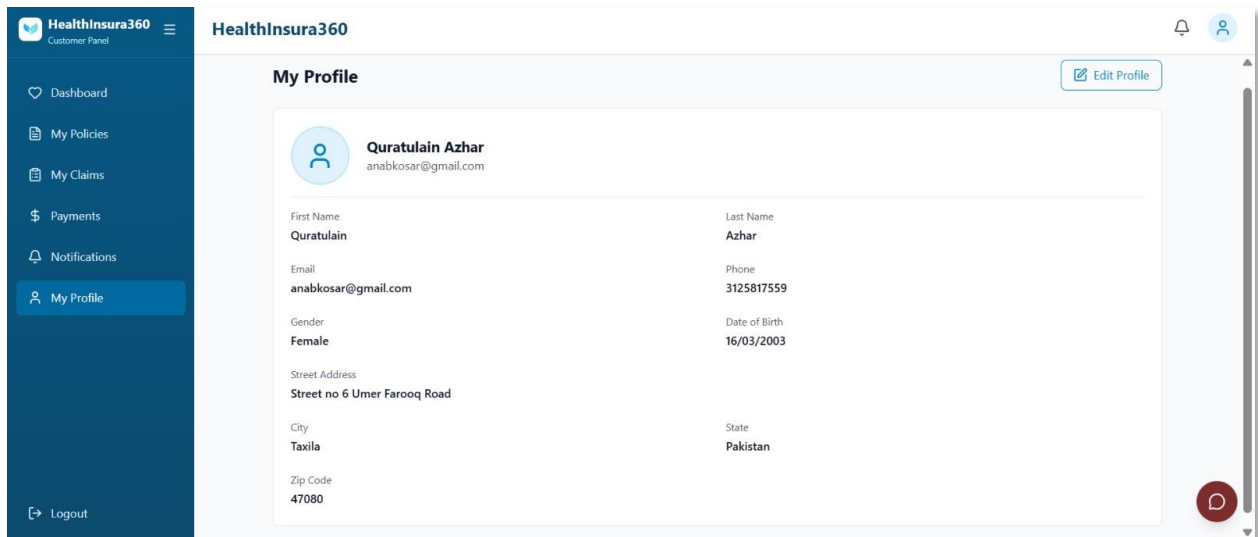
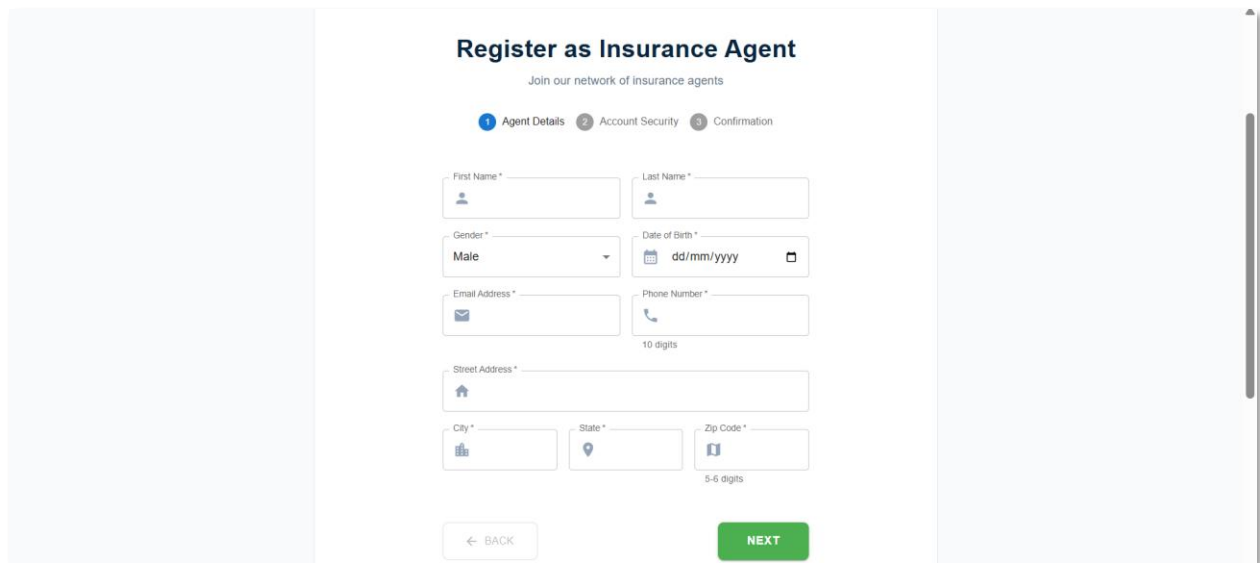


Figure 6. 9: User-Manage Profile

Manage Profile:

1. Click "My Profile"
2. View personal information
3. Click "Edit Profile" to update details
4. Upload profile picture
5. Change password if needed
6. Click "Save Changes"

6.3.2 Agent Operations



Register as Insurance Agent
Join our network of insurance agents

1 Agent Details 2 Account Security 3 Confirmation

First Name *
Last Name *

Gender *
Male

Date of Birth *
dd/mm/yyyy

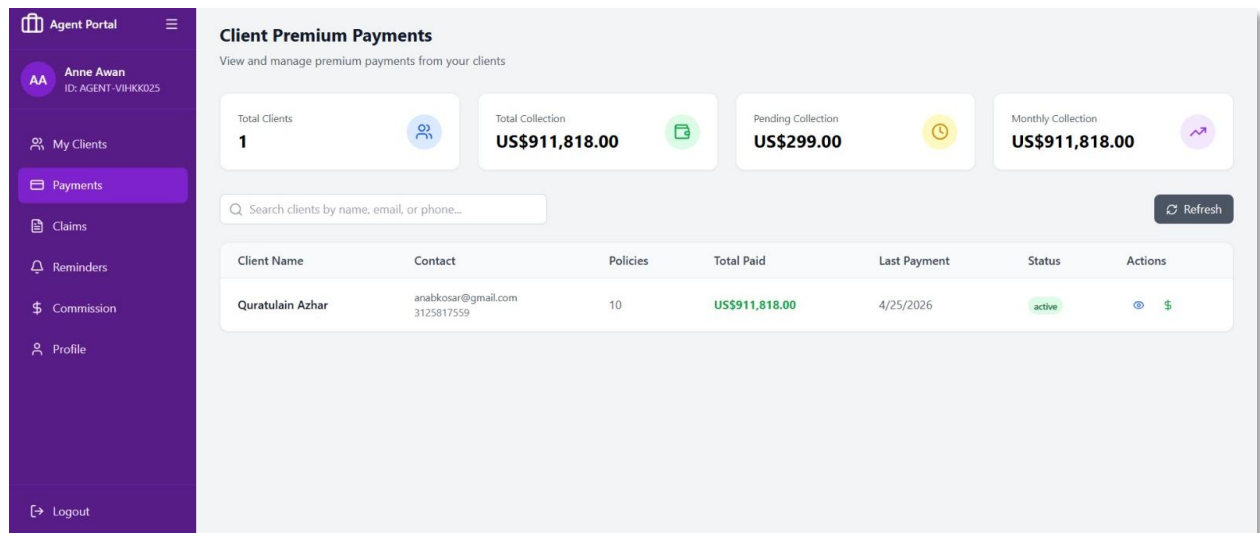
Email Address *
Phone Number *
10 digits

Street Address *

City *
State *
Zip Code *
5-6 digits

← BACK NEXT

Figure 6. 10: Agent-Registration Screen



Agent Portal
Anne Awan
ID: AGENT-VIH00025

My Clients
Payments
Claims
Reminders
Commission
Profile
Logout

Client Premium Payments
View and manage premium payments from your clients

Total Clients: 1
Total Collection: US\$911,818.00
Pending Collection: US\$299.00
Monthly Collection: US\$911,818.00

Search clients by name, email, or phone... Refresh

Client Name	Contact	Policies	Total Paid	Last Payment	Status	Actions
Quratulain Azhar	anabkosar@gmail.com 3125817559	10	US\$911,818.00	4/25/2026	active	👁️ 💵

Figure 6. 11: Agent-View Payments

View Payments:

1. Click "Payments" in the navigation bar
2. View all payment transactions

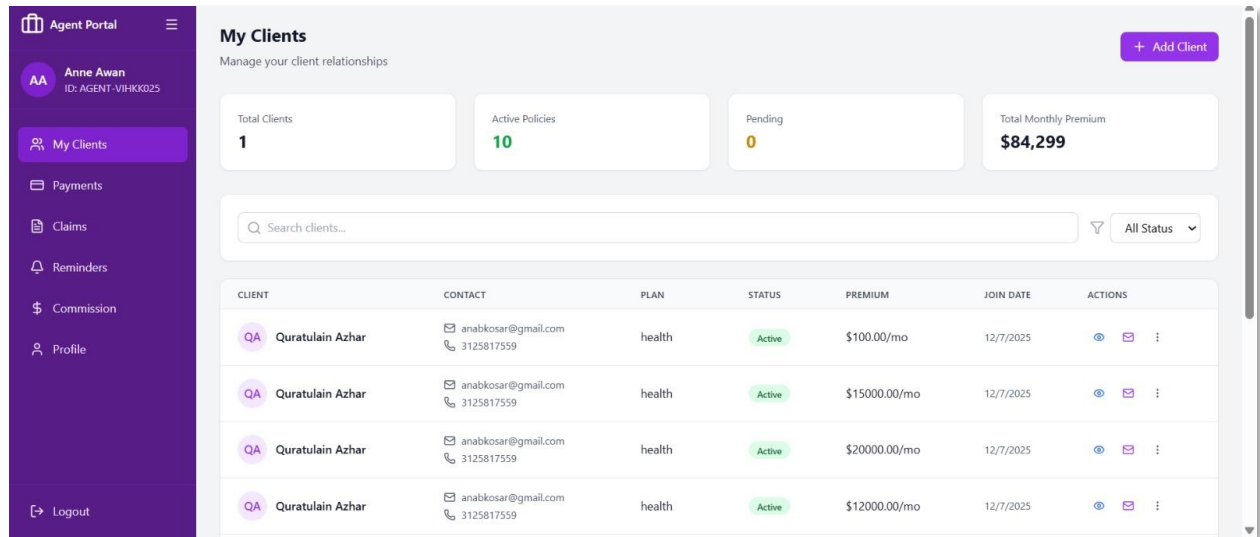


Figure 6. 12: Agent-View Customers

View Assigned Customers:

1. Click "Assigned Customers" in the navigation bar
2. View list of customers assigned to you
3. Click on any customer to view their details

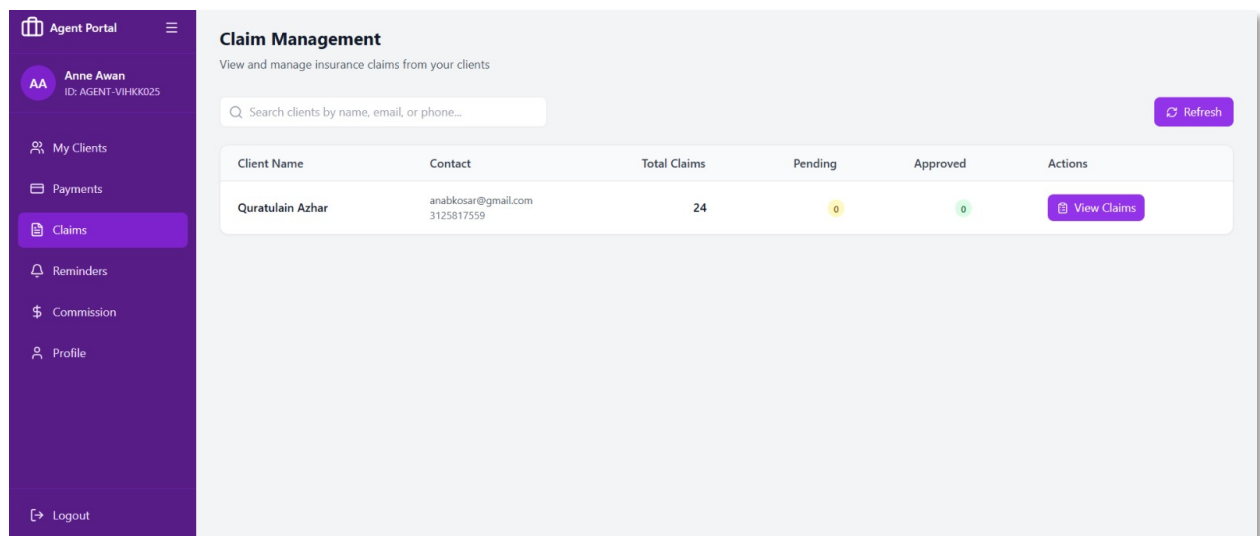


Figure 6. 13: Agent-Review Claims

Review Claims:

1. Go to "Claims Review"
2. View pending claims
3. Review documents
4. Click "Approve" or "Reject"
5. Add comments if needed

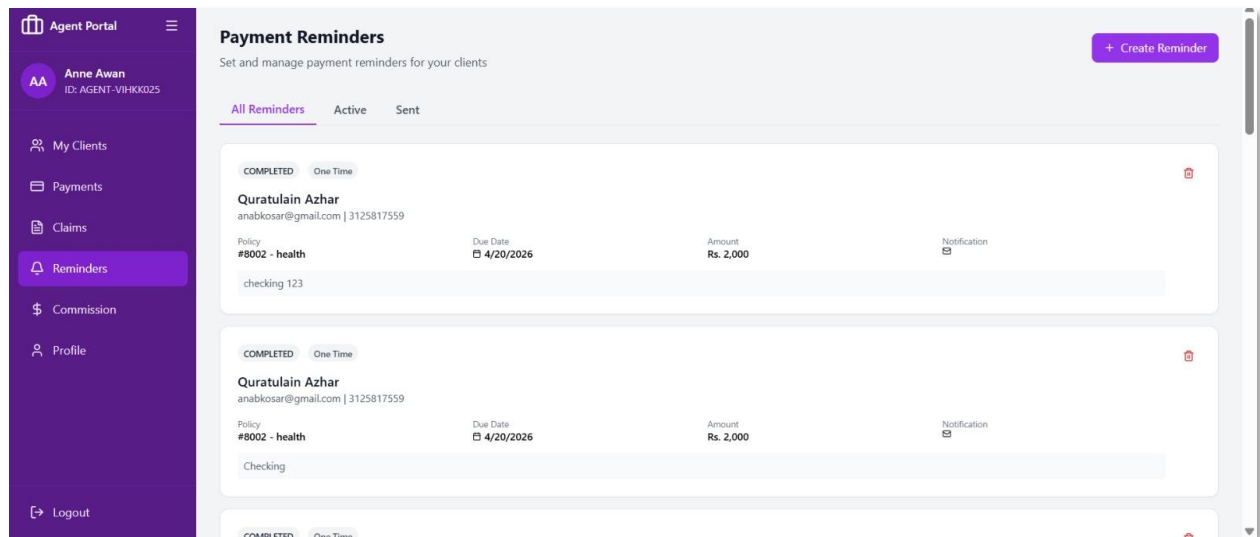


Figure 6. 14: Agent-Review Reminders

View Reminders:

1. Click "Reminder".
2. View recent Reminders

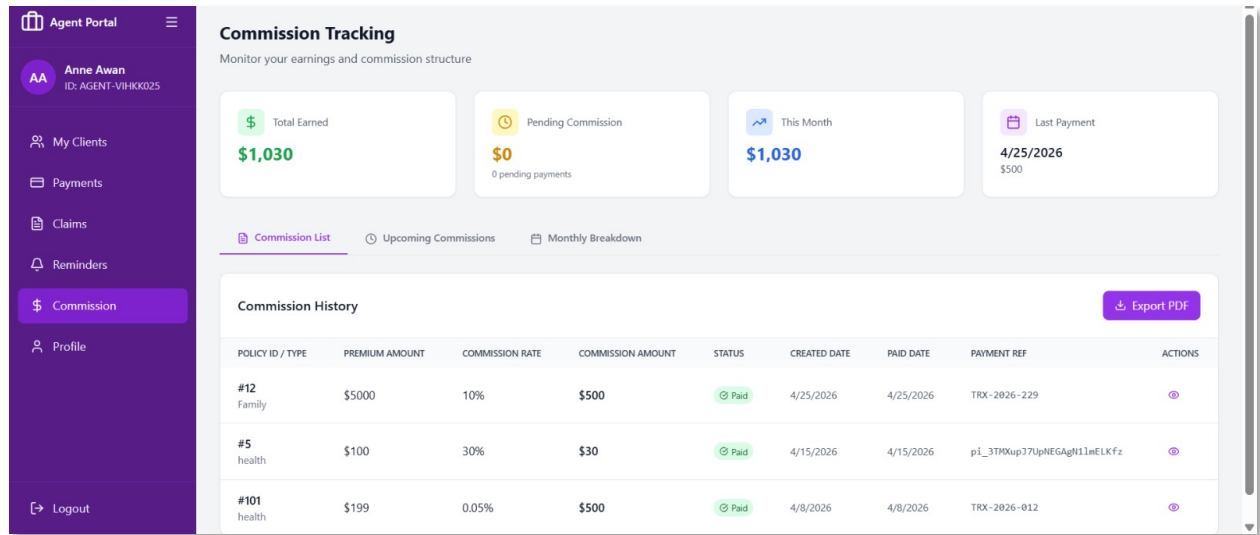


Figure 6. 15: Agent-Track Commissions

Track Commissions:

1. Click "Commission Dashboard" in the navigation bar
2. View total earned commissions

Agent Profile
Manage your profile information and credentials

Personal Information

First Name * Last Name *

Email Address
Email address cannot be changed

Phone Number

Office Address

[Save Changes](#)

Performance

Total Clients

Commission Rate

License Number

Member Since

Figure 6. 16: Agent-Manage Profile

Manage Profile:

1. Click "Profile"

2. View personal information (name, email, license number)
3. Click "Edit Profile" to update details

6.3.3 Hospital Operations

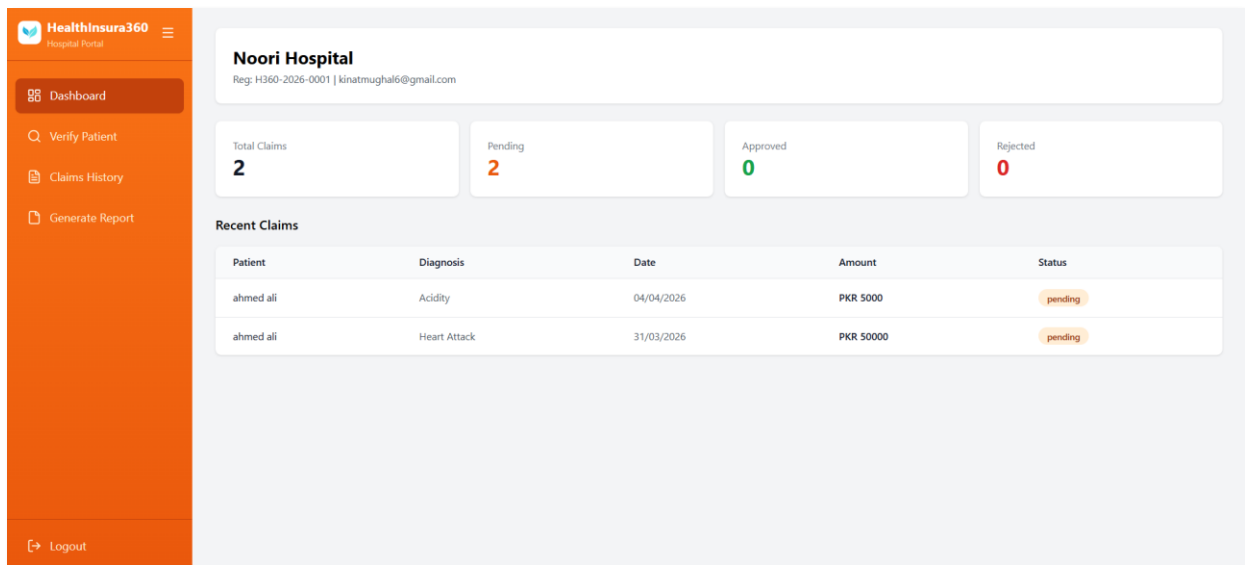


Figure 6. 17: Hospital-Dashboard

Figure 6. 18: Hospital-Registration Screen

Figure 6. 19: Hospital-Verify Patients

Verify Patient Policy:

1. Go to "Verify Policy"
2. Enter policy number or patient ID
3. View policy details

Patient	Diagnosis	Date	Amount	Status
ahmed ali	Acidity	04/04/2026	PKR 5000	pending
ahmed ali	Heart Attack	31/03/2026	PKR 50000	pending

Figure 6. 20: Hospital-View History

Download a comprehensive PDF report of all claims

[Download Claim Report](#)

Figure 6. 21: Hospital-Generate Report

Submit Cashless Claim:

1. Go to "Cashless Claims"
2. Select patient
3. Enter treatment details
4. Upload medical reports
5. Submit claim

6.3.4 Admin Operations

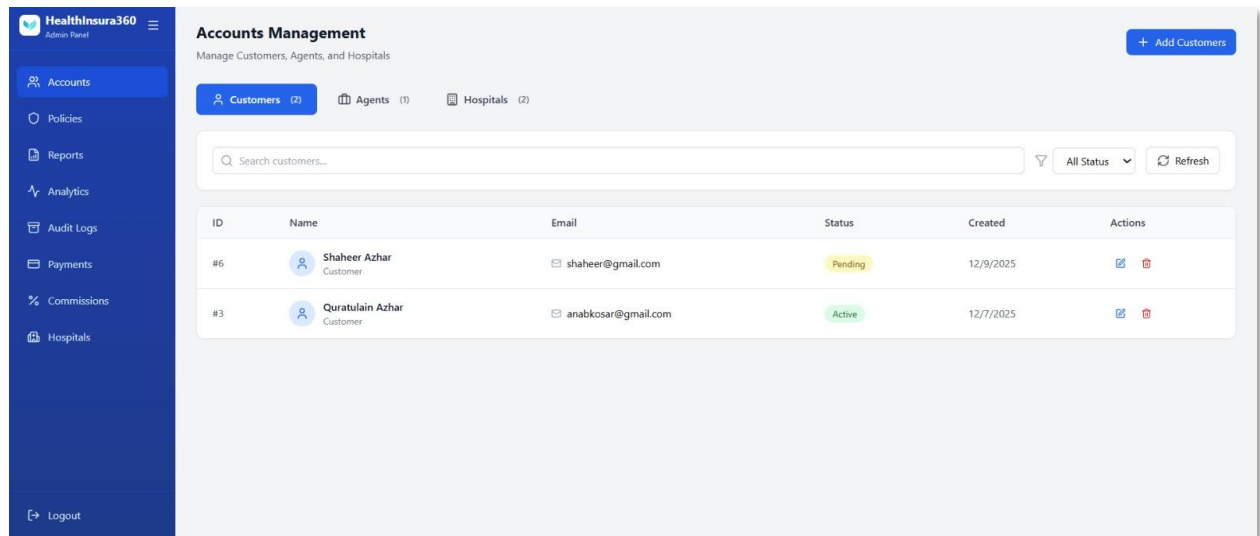


Figure 6. 22: Admin-Manage Users

Manage Users:

1. Go to "User Management"
2. View all users (Customers, Agents, Hospitals)
3. Click "Add/Edit/Delete"
4. Toggle account status

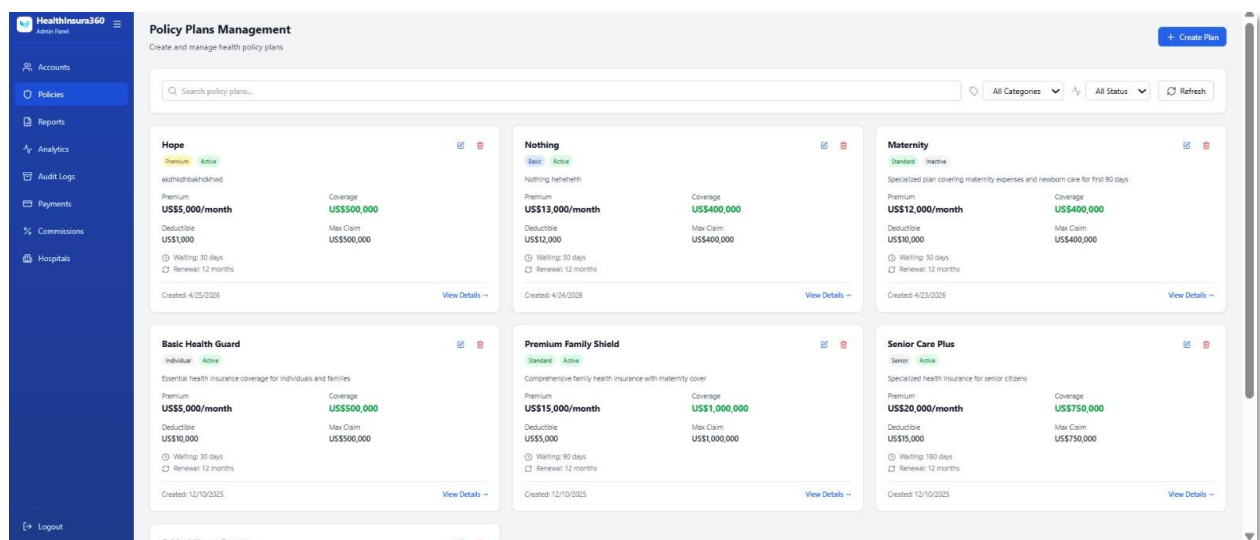


Figure 6. 23: Admin-Manage Policies

Manage Policies:

1. Go to "Policies"
2. Click "Add New Policy"
3. Enter policy details
4. Save changes

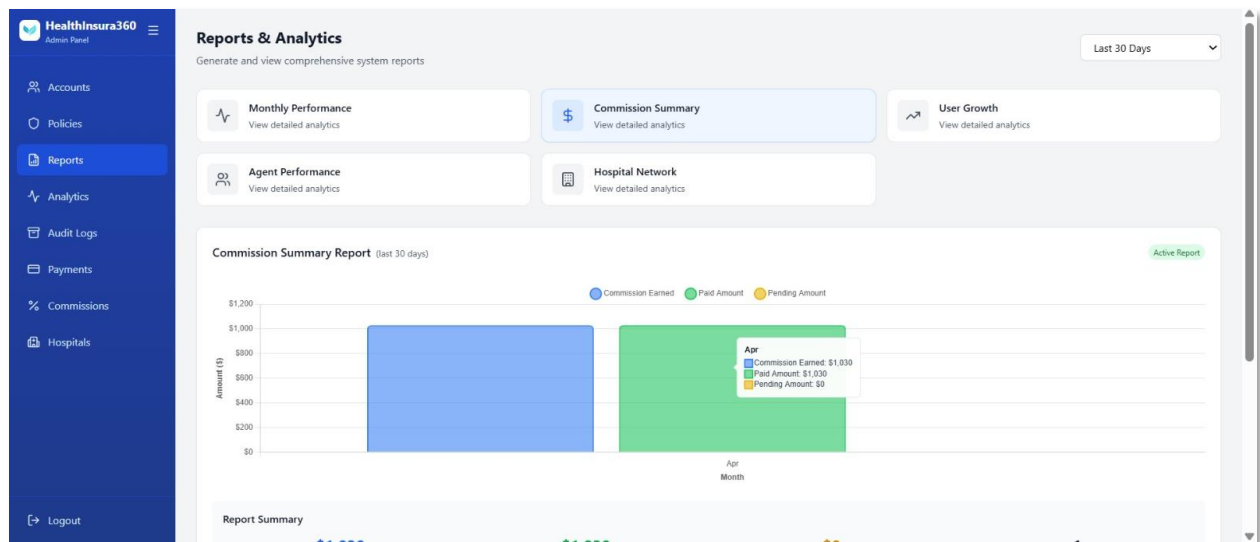


Figure 6. 24: Admin-Generate Reports

Generate Reports:

1. Go to "Reports"
2. Select report type (Claims, Policies, Payments)
3. Select date range
4. Download report (PDF/CSV)

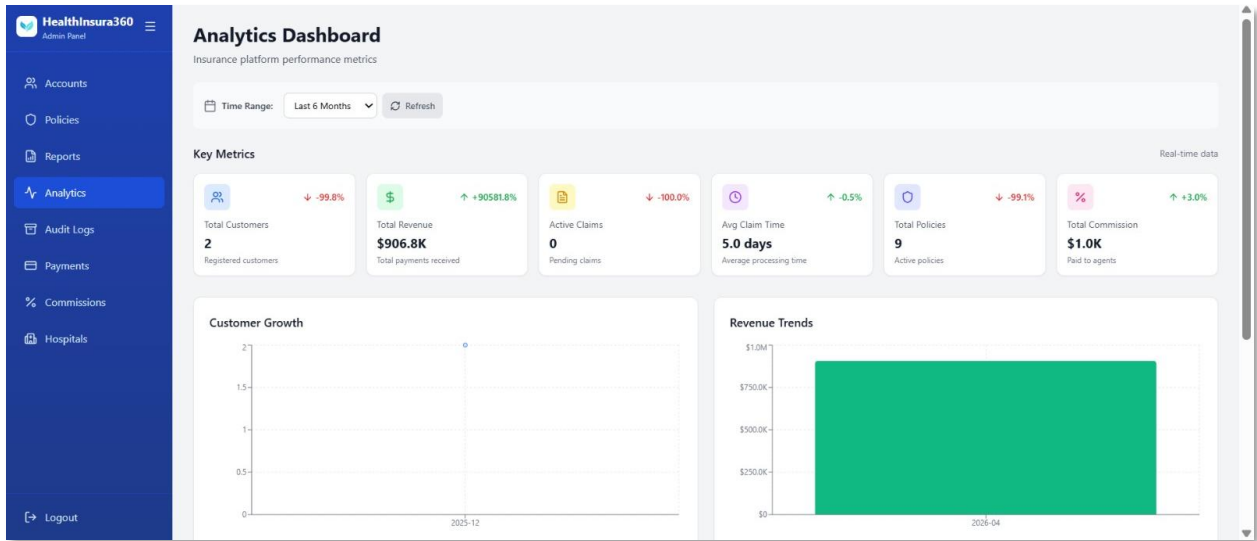


Figure 6. 25: Admin-View Analytics

View Analytics:

1. Go to "Dashboard"
2. View system metrics and charts

HealthInsura360 Admin Panel

Audit Logs
Monitor system activities and security events

Refresh Export Logs

Search logs... All Actions All User Types

Total Logs: 255

Info: 246

Warnings: 0

Danger: 9

Timestamp	User Type	User ID	Action	Entity	Severity	Details
Apr 25, 2026, 04:20:39 PM	admin	2	login	admin #2	Info	View Details
Apr 25, 2026, 03:11:04 PM	admin	2	UPDATE_HOSPITALS	hospital #2	Info	View Details
Apr 25, 2026, 03:07:26 PM	admin	2	UPDATE_HOSPITALS	hospital #5	Info	View Details
Apr 25, 2026, 11:35:40 AM	admin	2	login	admin #2	Info	View Details
Apr 25, 2026, 11:25:29 AM	customer	3	login	customer #3	Info	View Details
Apr 25, 2026, 11:25:09 AM	admin	2	CREATE_POLICY_PLAN	policy_plan #8	Info	View Details
Apr 25, 2026, 11:23:54 AM	admin	2	login	admin #2	Info	View Details

Figure 6. 26: Admin-View Audit Logs

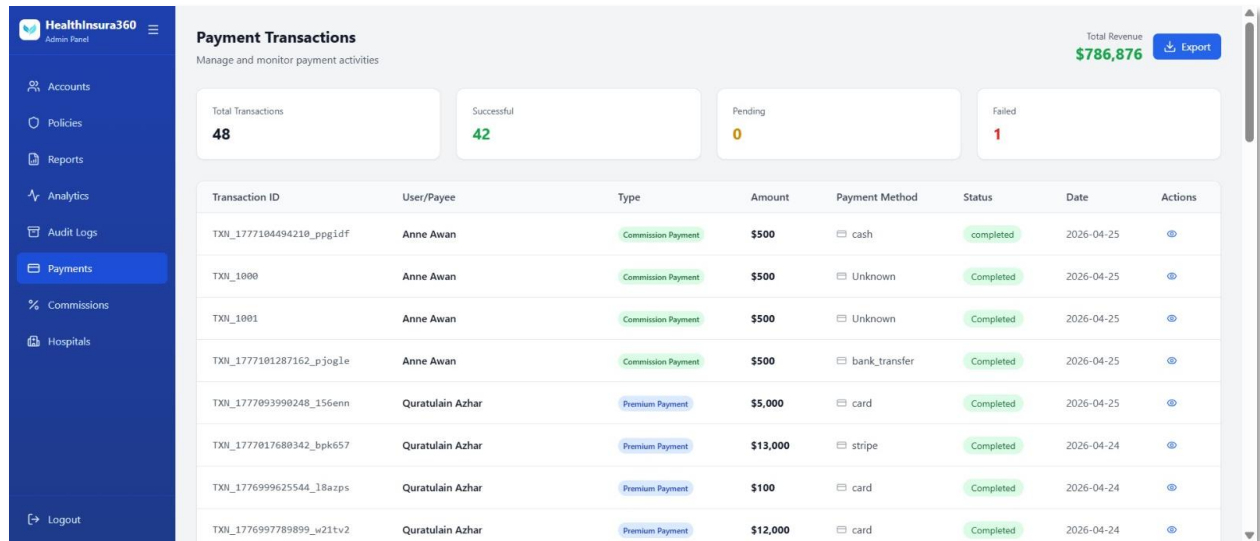


Figure 6. 27: Admin-View Payments

View Payments:

1. Click "Payment " in the navigation bar
2. View all payment transactions from all users

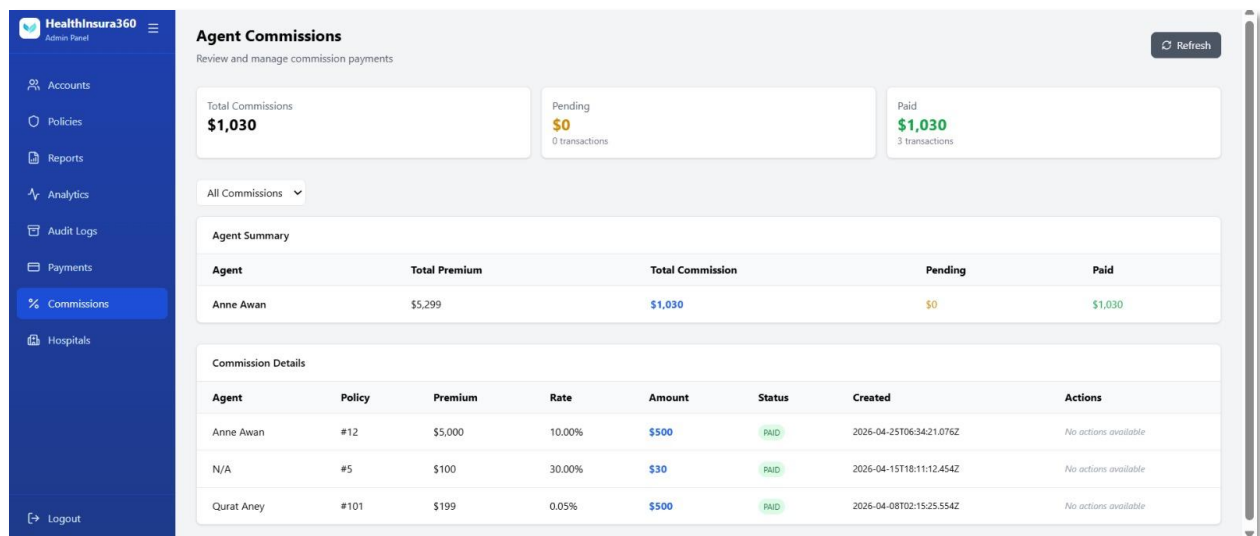


Figure 6. 28: Admin-Review Agent's Commission

Review Agent Commissions:

1. Click "Commission Management" in the navigation bar

2. View list of all agents
3. See commission earned by each agent

The screenshot displays the 'Hospital Network' section of the Healthinsura360 Admin Panel. The interface includes a sidebar with navigation options: Accounts, Policies, Reports, Analytics, Audit Logs, Payments, Commissions, and Hospitals (highlighted). The main content area features a header with the title 'Hospital Network' and a subtitle 'Verify and manage network hospitals'. A '+ Add Hospital' button is located in the top right corner. Below the header, there are four summary cards: 'Total Hospitals' (2), 'Verified' (1), 'Pending' (1), and 'Active' (0). A table lists the hospitals with columns for Hospital Name, Location, Contact Person, Email, Status, and Actions. The table contains two entries: 'Noori Hospital' (Verified) and 'POF' (Pending). The 'Actions' column for each entry contains icons for edit, view, and delete.

Hospital Name	Location	Contact Person	Email	Status	Actions
Noori Hospital	Wahcantt, Pakistan Basti	Noor	kinatmughal6@gmail.com	Verified	
POF	WahCantt, Pakistan Main Cantt Road, Near Barrier no 1	N/A	pof@gmail.com	Pending	

Figure 6. 29: Admin-Manage Hospitals

7: CONCLUSION

7.1 Summary

HealthInsura360 successfully delivers an AI-powered health insurance management platform that revolutionizes the traditional insurance experience. The system integrates policy comparison, AI-driven recommendations, claim processing, and secures digital payments into one unified ecosystem.

7.2 Achievements

The project achieved the following:

- i. **Complete User Management:** Secure registration, login, and profile management with JWT authentication.
- ii. **AI-Powered Features:** Personalized policy recommendations and intelligent chatbot for real-time assistance.
- iii. **Digital Claim Processing:** End-to-end claim submission, tracking, and approval of workflows.
- iv. **Multi-Stakeholder Support:** Specialized portals for Users, Agents, Hospitals, and Admins.
- v. **Secure Payments:** Integration with payment gateways for policy purchases and renewals.
- vi. **Analytics Dashboard:** Comprehensive reporting and system monitoring tools.

7.3 Future Work

Potential enhancements for future versions include:

- i. Mobile application development for iOS and Android
- ii. Blockchain integration for immutable claim records

- iii. Advanced fraud detection using machine learning
- iv. Telemedicine consultation integration
- v. multi-language support
- vi. Voice-enabled chatbot
- vii. Integration with government health schemes
- viii. Biometric authentication